Engineering Chemistry · Unit_04 · Functional materials and green chemistry

DRAFT — UE26CY151A · Unit_04 — Functional materials and green chemistry · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_04 · Functional materials and green chemistry

1 UE26CY151A · Unit_04: Functional materials and green chemistry

This unit master organizes source material for Engineering Chemistry, Unit_04, officially titled Functional materials and green chemistry. It includes 12 video transcripts and 12 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26CY151A · Unit_04: Functional materials and green chemistry

Official catalogue entry: UE26CY151A, Unit_04 (Functional materials and green chemistry). Bucket folder present: yes. Videos: 12; slide decks: 12.

EV0000001

2 Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties -blue shift (Part 2)

Study focus: Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties -blue shift (Part 2). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module two on nanomaterials and in this class we are going to talk about properties of nanomaterials, surface area dependent properties, electrical properties and optical properties.

Key details

- Properties of nanomaterials are very different from that of the bulk materials.
- We find that for bulk material the properties are independent of size whereas when we come to nanomaterials they change with size.
- Then it is it has a particular colour that is called golden colour.
- There is no change in the colour, it still continues to remain in that golden colour.
- But that is not true when we come to the nanoparticle range.
- Gold, when we get it into the nanoparticle range, we find that as we decrease the size, the colour of gold also.
- That the optical properties are getting altered with the dimension of the particles.
- Is what happens for all most of the properties of nanomaterials.
- They change with size.
- Whereas in nanomaterials, the material remains the same.
- The structure composition remains the same, it's only the size which is changing, and that can alter the properties.
- We know that many physical and chemical properties of a material are going to depend on surface properties, and when we come to nanomaterials, the surface area has enormously increased.
- We have a bulk material over here, it is being broken down into smaller particles.
- We know that the surface area has increased so much over here.
- The surface area.
- And we know that since in nanomaterials the surface atoms are much more than what the atoms at inside the bulk of the material and therefore.
- All the properties are mostly dictated by the surface atoms.
- And since we have such a large surface area, such a large number of surface atoms, therefore the properties which depend on surface area definitely get affected when you take.
- the particle when take the material from bulk to nanocyte.
- So properties like catalytic activity we know that whenever we use a catalyst we prefer to take it in the powder form because we want to increase the surface area and catalytic catalysis is a surface phenomena.
- If we take it in nano range, nano particle range, the catalytic activity is going to increase enormously.
- Are found to be catalytically very active for selective redox reactions.
- Because there are a lot of surface atoms which are held much less tightly as compared to the bulk atoms.
- For nanoparticles, it is found that electrical conductivity decreases as compared to the bulk material.
- One important reason for that is spatial confinement.
- We talked about spatial confinement also, and we understood that as we go into the nanoparticle range.

Example or application from the lecture

- For example, if we have a block of gold.
- Now even if you make it into smaller and smaller size, like for example, the block of gold which was say hundred grams, now we make it into a five gram piece, by breaking it into smaller pieces, we find that.
- Let us look at the most important property of nanomaterials, the surface area.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

The energy levels become more and more discrete, and the band gap increases or the energy gap increases.

Key details

- What are we talking about here?
- Now when we have metals, we know that the valence band and the conductance band are overlapping and therefore the.
- Electrons can easily move into the conduction band and we find that metals are good conductors.
- Whereas in insulators the energy gap is very high and therefore they are not found to conduct electricity.
- Whereas in semiconductors the band gap is less.
- Energy gap is less now, and therefore, it is possible that if we give the right energy, the electrical conduction is possible.
- You can see the same energy.
- Semiconductor now the band gap has become more and obviously when the band gap is more than the conductivance properties are going to get altered.
- you can see here this the band gap is so high, so obviously the conductance has become less and they can almost be acting like insulators.
- Another important reason is due to surface scattering.
- What is a mean free part for an electron?
- the distance which the electrons move before they undergo some collusion.
- We know that the electron has a mean prepared and there's no obstruction.
- But when we have the same bulk material into the nanoparticle range, now we know that the nanoparticle has become has a very small dimension.
- Changing its direction and therefore there is conservation of energy as well as momentum.
- Whereas if it is inelastic scattering, then when the electron is scattered in randomly in all directions.
- We find that the scattered electron now loses its velocity, and when the electron loses its velocity, it's understood that the.
- Mobility is now less and therefore the electrical conductivity decreases.
- Coming to the next property that is optical properties.
- Nanomaterials show unique optical properties.

- As I mentioned, the colour depends on the size of the particles.
- In the bulk material, there is no change in colour even if you decrease the size.
- But when you come into the nano range, it is found that as we change the size of the particles, there is a change in colour.
- One is due to the surface plasmon resonance and the other is due to increase in band gap.
- If you look at metals.
- This is the picture of a metal.
- What happens is the.
- Electric field which is fluctuating, the electromagnetic radiation which is formed falling on the surface, it has an electric field.

Example or application from the lecture

- Let's understand what is the surface plasmon resonance.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

Because the lattice point is positive, these is an electron cloud which is getting formed, there is an attraction which is going to pull these electrons back.

Key details

- Up in the in the system.
- This collective oscillation of these electrons is what is known as the plasmon.
- Collective oscillations of these electrons is what is known as plasma.
- Then there is going to be resonance.
- When resonance happens, we know that the radiation is going to be absorbed, and when radiation is absorbed in the visible light, then the material appears to be colored.
- This is going to depend on the size, shape and nature of the metal.
- Now it is not just the just this happens because there's an electron cloud and a positive metal side.
- It is not the same for all the metals.
- It is going to depend on the nature of the metal, it also depends on the size of the particle.
- Why is it dependent on the size of the particles?
- Because we know that the electric field there is a polarization.
- This electric field is definitely going to depend on the size, not only on the size, it also depends on the shape.
- Whether the particle is spherical or whether it's rod shaped or it's a platelet shape, then definitely the.
- Electric field is going to be different each time, and therefore the plasma frequency is different.
- And if the plasmon frequency is different, definitely the light absorbed is going to also be different.
- Therefore, as the size of the particle changes, the color of the.

- The nanoparticle also changes.
- Another reason is the increase in the energy gap.
- This is again what we have understood before as spatial confinement or quantum confinement.
- We know that as we go into the nanoparticle range, when we go to smaller and smaller sizes, the energy levels become more discrete and the energy gap increases.
- Now if energy gap increases, obviously the wavelength of the light absorbed is going to be different and therefore.
- As we increase as the energy gap goes on increasing, then the energy required is going to be more, so the wavelength is going to be less.
- And that is exactly the reason why we find that as the size of the nanoparticles becomes.
- Smaller and smaller, we find a blue shift which happens in the in the particles.
- Is decreasing the energy gap increase.
- This we understood in when we were talking about spatial confinement and the wavelength of the light absorbed, as I mentioned just now, goes towards smaller values, and this is referred to as blue shift.
- Suspensions of gold nanoparticles, as you can see here, as the size is increasing, or if we say as the size is decreasing, we find there is a change in color.
- This is very characteristic of nanoparticles.
- Nanoparticles gold spheres which have greater than 50 nanometers they absorb at a wavelength of 575 nanometers now gold spheres 10 to 20 nanometers absorb at 524 nanometers whereas 2 to 5 nanometers absorb at 517 nanometers what does it mean?
- The wavelength has become less, that means the frequency is more, that means the energy gap is more.
- As we decrease the size, we find that.
- the energy gap is becoming more and more and the light now is now moving the wavelength is now moving more towards the violet so we call it as the blue shift.

Example or application from the lecture

- So let us take the example of gold.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

So in the next class we are going to talk about some more properties of nanomaterials like the mechanical properties, the thermal properties, and magnetic properties.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

3 Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties -blue shift (Part 1)

Study focus: Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties -blue shift (Part 1). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000018 · EV0000019

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will start with a new module today on nanomaterials.

Key details

- The content will be nanomaterials, classification and deviation of properties from bulk material, and we will look at different properties of nanoparticles.
- In today's class we will study of nanomaterials, about nanomaterials, the classification of nanoparticles, and the reasons for deviation in properties of nanomaterials from the bulk material.
- The prefix nano is derived from the Greek word dwarf meaning very tiny.
- One nanometer would be approximately equal when we have ten hydrogen atoms placed one next to the other or five silicon atoms.
- aligned in a line.
- Previous module we studied about molecular spectroscopy and we understood that when we study about molecules or atoms we need to use the quantum mechanical treatment.
- But when the dimension becomes close to the nanometer, then we are going to observe.
- Because the dimension are now, dimensions are closer to those of atoms or molecules.
- because their properties are very different from that of the bulk materials.
- All of us know that bulk gold as we see it is yellowish in colour.
- That means there is some change in the optical property.
- That means the activity has become so high something involved which is passive has become so active that it is combustible to the extent that it is used as a solid fuel in rocket propulsion.
- Another property of importance is melting point of the materials when they are taken in the nanoparticle range becomes lower.
- If we want to check purity of the material, we always quote the melting point.
- This material will be classified as a nanomaterial.
- And what would be nano science?
- It is the study of the phenomena and synthesis, characterization, exploration, and exploitation of these nanostructure materials.

Example or application from the lecture

- Let us first understand what we mean by a nanoparticle.
- For example, gold.
- In fact the a name golden colour has been given to gold.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

Core explanation · 05:00–09:59

Before studying the reasons why nanomaterials behave so differently from the bulk material, let us talk about classification of nanomaterials based on their dimensions.

Key details

- Now the first classification would be it will be called as a zero dimensional nanomaterial.
- What does it mean?
- Range exceeding 100 nanometers.
- It's understood now that two of the dimensions are allowed to be outside of the nanorange.
- films these will be classified as two dimensional nanomaterials only one dimension is now within the nano range but still we call it as a nanomaterial.
- And then coming to three dimensional nanomaterials where all the three dimensions are outside of the nano range.
- Then how is it a nanomaterial only specific types of materials where you have like say bundles of nanovires.
- Nanomaterial, therefore, these overall will look like as if all the three dimensions are outside of the nanorange.
- First and foremost reason which we should discuss here is that the large fraction of surface atoms per unit volume.
- Atoms on the surface, but major of the atoms are in the bulk of the material.
- But when we come to nanomaterials, what happens?
- The particle size has.
- Become so small now that on the surface you have a large number of surface atoms, whereas in the bulk we have very few atoms now.
- Properties now are strictly governed by the surface atom because they are larger in number as compared to the bulk atoms.
- When we have a platinum sheet as compared, which is we can say is bulk, whereas compared to platinum powder, but to a nanopowder.
- Now the surface area, each particle now, the surface area has is more, and therefore overall surface area.

Example or application from the lecture

- Example, as I said, nano clusters or quantum dots fall into this category of nanomaterials.
- An example would be nanotubes and nanovires we can understand look at the picture here.

- So such materials are also classified as three dimensional nanomaterials where we have all the three dimensions outside of the nanorange like for example dispersions of nanoparticles or we have bundles of nanovires.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 10:00–14:59

has become so high that the number of surface atoms are very large and therefore we will find that the properties now will be dictated by these atoms rather than the bulk atoms.

Key details

- We have a box like this.
- Dividing it further and further, the surface area is definitely going to increase and the number of surface atoms will go on increasing.
- We all know that the bulk atoms are surrounded by atoms on all sides and therefore there's a force of attraction bit and repulsion between these bulk atoms.
- But when we look at the surface atoms they are being held or pulled inside or held together with only one on only one side that other side other side is exposed to the environment therefore there is they are held with less force and therefore there is more energy on the surface and when any.
- System has more energy tries to it tries to reduce its energy and therefore we'll find that the surface activity will be very high.
- Of nanoparticles, we find that the same aluminium becomes so combustible that it is being used as rocket fuel.
- So therefore this.
- The reason being that there are large number of surface atoms and they are being held with less force, so surface energy is very high.
- Another reason which is very important and if we understand this, all properties, especially electrical and optical properties, can be explained very easily.
- by the use of the concept of spatial confinement or quantum confinement.
- We have these metal at the lattice points and we have a sea of electrons.
- That is for a bulk.
- You make it so small and so thin that it has now become into a two dimensional material.
- What would happen?
- when you make it into a two dimensional quantum well say.
- Your electrons can move only in one direction.
- In the other two directions, they are completely confined.
- You have changed, you have made it so thin that now it is allowed to move only in one di dimension.
- In the holes to move only in one direction, and when you come to zero dimension, which are called quantum dots, you have restricted the motion completely, and it's almost like within a very small area, you are confining.
- The electron and the poles.

- When we have a bulk material, we know that.
- Say it is one hole of the material or the metal we are considering.
- One hole would mean to the 10 to the power of 23 atoms.

Example or application from the lecture

- And just now I gave an example of aluminium where we find that aluminium in bulk is passive in nature, whereas when it is taken in the form of.
- We know that let us take the example of a metal.
- But now suppose you make this bulk material so thin, so thin, so thin.

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EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 15:00–19:59

compared to nanomaterials where we'll have a maximum of ten to thousand atoms.

Key details

- Now here, when we say one mole, that means we have 10 to the power 23 atoms, that means so many electrons, and also means that there are so many energy levels.
- They'll be very close and almost continuous.
- They will be continuous to the extent that in metals we know that the balance band and the conductance band are overlapping.
- When we come to semiconductors, of course there is a small gap, nevertheless, these levels and the levels in the valence band and the condensed band, there are so many levels because there are so many atoms.
- And these levels will be very close to one another that they will be continuous.
- But when we come to nano range, what happens is now we have reduced the number of atoms.
- There are fewer atoms, fewer electrons, so the density of states falls down.
- And what we observe, say we are in a two dimensional.
- quantum well then we'll find that the energy levels now because there are less number of atoms, less number of electrons, less number of energy stays.
- And then when we come to zero dimensional nanomaterial the energy levels now are completely discrete and the energy gap has increased.
- If you compare them with a molecule in a molecule we know we have absolutely discrete energy levels.
- to the lumino, the highest occupied molecular orbital to the lowest unoccupied molecular orbital, and this is the energy gap.
- But as these become more and more in number, if you go this way now.
- When we come to bulk, we find that the valence band, there are so many.
- Many energy levels now because of so many atoms there.
- or the spatiment.
- What do we understand?

- That due to spatient or quantum confinement, the energy levels become more and more discrete and the energy gap increases.
- This is the result of quantum confinement, and this is going to have a strong bearing on the properties, especially the electronic properties, electrical properties, and the optical properties.
- And one more reason.
- for the bulk material behaving differently from the nanomaterial, but the nanomaterial behaves different from the bulk, is that in the bulk material we find that there are lattice imperfections like wherever there are dislocations, kinks, etcetera, are found in the lattice.
- Because the metal lattice is formed in such a way that layer after layer after layer is getting deposited of the metal.
- And many times we do not get a perfect layer.
- Rather there are some imperfections which are like dislocations or kinks are possible.
- in the metal structure, it's a bulk metal structure.
- But when you come to the nanomaterial range, what has happened now you just have to they're very small in size the particles.
- Therefore very small lattice has to be formed and therefore there is less possibility of imperfections.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 20:00–24:59

And in the next class we are going to study how exactly these reasons are affecting the thermal properties, optical properties and so on so forth of the nanomaterials.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

4 Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials

Study focus: Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000038 · EV0000039

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module two on nanomaterials and this will be the last class of this module.

Key details

- And finally, we will look at some applications of nanomaterials.
- When we talk of mechanical properties, the most important thing we need to understand here that the strength of the nanomaterials is found to be greater than the bulk material.
- What does it mean?
- When we take a substance in the form of a bulk and it has a certain strength.
- But when you come to the nanomaterial range, it is found that the strength increases to a great extent.
- Bulk copper we know can be bent with certain amount of ease.
- We take these copper particles in the nano range, like when you get it to below 50 nanometers, it is found that they become super hard in nature.
- The same copper which in the form of bulk could be bent cannot.
- We bend now and it becomes super hard and it loses all its malleability and deductivity, which we find in the case of bulk copper.
- The reason is that when you take a particle in the nano range, the copper.
- in the nano range the number of defects becomes very less there is less probability of defects occurring now what do we mean by this.
- When we have large lattice lattices as in the case of bulk material, we know one layer above the other is getting formed when the metal is being when we have a metal.
- There are one after the other layers are getting formed.
- In these layers we will find that there could be imperfections like dislocations, kinks, and vacancies.
- And because of these imperfections or these vacancies and dislocations, etcetera, we find that the strength of the material.
- [ASR low confidence; check audio] material becomes less, but when we.
- Have particles in the nano range, we know that we are forming very small lattice which has a smaller cross section, and therefore the probability of finding these imperfections is very less, and therefore we find that when we go to this nano range of such a small particle.
- Of less than 15 nanometer, definitely the number of imperfections is less.
- The first property we need to discuss is the melting point.
- What would be phase transition temperature when you change from solid to liquid.
- That temperature is also reduced, or other phase transitions could also happen with two different types of solids.
- Also, the same solid existing in polymorphic forms that also would be a phase transition.
- Point also decreases.
- They are very characteristic.
- In fact, melting point, if you want to check the purity of a metal the fur or of any material, the first thing we do is check the melting point.
- And that would be characteristic of the material.
- And we know if the material if it is showing this melting point.
- Point which is listed, then we know that the material we have taken is pure.

- But the same material in the nano range now the melting point becomes less.
- What is the reason for that?
- It's very simple to understand.
- We know that.
- There are a lot of surface atoms in the nanomaterials because the particles are so small, most of the atoms are on the surface.

Example or application from the lecture

- Now let us take an example of copper.
- We have a very good lattice, and that results in making copper, for example, a super hard material.
- Let us talk of thermal properties.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 05:00–09:59

Obviously, you need to break the bonds.

Key details

- Now these surface atoms are held with less energy, so less energy will be required to break these bonds, and therefore we have lesser bonds, so the less energy.
- energy will be required to break these parts and hence we find that the melting point is going to decrease.
- So because of these large number of surface atoms, we find that the melting point of materials in the nano range becomes less.
- Another property which changes is the thermal conductivity.
- It is found that there is a decrease in thermal conductivity.
- It is due to the vibration of these lattices.
- obviously it will result in scattering which is known as phonon scattering and once this scattering happens that means it is going to result in a decrease in the thermal conductivity.
- We find that the magnetic properties of nanomaterials are distinctly different from that of the bulk materials.
- You take a material in the bulk form, and when you go to the nano form.
- Range of the same material, the magnetic properties are found to be very different.
- Iron cobalt and nickel.
- But the iron cobalt and nickel when they are taken in the nano range, we find that this ferromagnetism changes to superparamagnetism.
- in the nanometer range.
- The magnetism doesn't come to zero when we remove the magnetic field in the case of a ferromagnetic materials.
- But the same material now I don't when it is taken in the nano range.
- We find that they change into paramagnetism.
- They shall start showing paramagnetic behavior.

- The material gets aligned in a magnetic field but to a very small extent.
- But in the case of nanomaterials.
- The material which was ferromagnetic does become paramagnetic, that means it loses its magnetism to a great extent.
- but in the presence of the magnetic field we find that the magnetic susceptibility is very high.
- Because we know that the surface energy is very high.
- Because the surface energy is high, now these magnetic dipoles they can flip.
- These will flip in the in the presence of the magnetic field, they will they can flip at room temperature itself, they can flip directions and therefore they lose their ferromagnetic nature.
- But nevertheless, when there is a magnetic field, they will get.
- Magnetized like a paramagnetic material.
- But the magnetization is going to be very high, so that the magnetic susceptibility is going to be very high in this case.
- And when you remove the magnetic field, they come back to zero.

Example or application from the lecture

- Let us talk of magnetic properties.
- Let us take up the example of.
- Let us first understand what exactly is a ferromagnetic material from this graph and what is a paramagnetic material so that we can understand what is that change.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 10:00–14:59

behavior so from ferromagnetic it changed to superparamagnetic behavior why the reason is the high surface energy so the domains at room temperature itself flip directions and therefore they lose their alignment with the ma alignment and therefore when the magnetic field is applied they do get some small amount of alignment like paramagnetic behavior but the susceptibility is now very high.

Key details

- They get magnetized to a great extent and that is called superparamagnetic behavior but the g the interesting part is that when the magic field is removed they don't behave like ferromagnetic beh material which can retain they have memory but here it becomes zero.
- They are non-magnetic, but then in the nano size range they start behaving like magnetic particles.
- We know that nanomaterials have a range of applications in many fields.
- Then energy storage, catalysis, then consumer electronics, environmental and cosmetics.
- And then we can also have injectable nanopods which can serve as detectors, informers of early disease and perhaps repair metabolic defects.
- And apart from that we know that the there's.
- Silver nanoparticles are known for their antibacterial activities.

- There are several more applications in medicine.
- The region of energy storage.
- The electrodes which are used in fuel cells.
- The electrodes need to have very high.
- surface area because that is where the actual reaction takes place.
- The electrode is in contact with the fuel or the oxidant, it is also in contact with the catalyst.
- The catalyst being is being impregnated onto the electrode and it's also in contact with the electrolyte.
- Therefore a high surface in the or area of the electrode which would help in making the fuel cells very efficient.
- Then coming to catalysis.
- Now nano crystalline materials again they have a very large surface area because they are in they have a very small size and most of the atoms we find on the surface and these atoms will also be very active so with regard to their physical, chemical and mechanical properties, so we know that they can serve as very good catalysts.
- If we use the material in as a nano particle.
- In the nanoparticle range, we find that they will serve as a very good catalyst.
- And in fuel cells, again, oxygen reduction reaction is a very slow reaction.

Example or application from the lecture

- Now another property is the another example is where we have bulk, gold and platinum.
- For example, when we have these cancer cells and some drugs are given, we know that the they don't not only affect the cancer cells, but the cell around are also getting affected.
- For example, we know gold and silver nanoparticles are being used as catalysts in several experiments.

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EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 15:00–19:59

while make for very good catalyst and also as a catalyst for carbon monoxide oxidation.

Key details

- They are composed of materials which are known as phosphors which glow when they are struck with a stream of electrons from the within the catchment rate two.
- Size or the phosphorus.
- So obviously, if we take nanophosphors, then the size is sufficiently reduced, and therefore we find the number of pixels is more, and therefore we will find that the resolution is going to be much higher.
- materials like zinc selenite, then cadmium sulphide, zinc sulfide and lead sul tellurite.
- And nanophosphors are also being employed in white light emitting diodes.
- Nanomaterials can be very useful and are being employed in catalytic converters.

- Now because they have high chemical activity and due to the high surface area, therefore, nanomaterials are being used as catalysts to remove the toxicity.
- And the noxious gases such as the nitrogen oxide and carbon monoxide in automobiles as well as for power generation equipment.
- no, earlier when sun blocks used to be used they would be white in colour.
- But now because.
- Of the use of these nanomaterials, now these same zinc oxide and titanium oxide, they are taken in the nanoparticle range, and it is found that they now because of the nanoparticle range they become transparent.
- out you don't get that whitish color on the face and they are now transparent.
- These are just some of the applications which we have listed over here but yes they have a several.
- Applications in the electronics industry and or many applications are found for nanomaterials.
- So in the next class we are going to start with unit two where we will study phase equilibrium.

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EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

5 Polymers: Introduction, types of polymerizations, definition of glass transition temperature

Study focus: Polymers: Introduction, types of polymerizations, definition of glass transition temperature. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000054 · EV0000055

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we'll start with a new module on functional materials.

Key details

- In that module we are going to study about polymers, the types of polymerization, structure property relationship, commercial polymers will take up butile rubber, biodegradable polymers, epoxy resin, polymer composites, cablar and carbon fiber.
- Conducting polymers and we will look at molecular weight of polymers.
- Then we'll also study organic LEDs.
- First, we will take up some definitions.
- A giant molecule or a macromolecule, which is formed by repeated joining of several simple molecules.

- If you look at the picture here, we can see some small molecules over here and they have been joined together by a process which is called polymerization to give this long chain and this chain of these simple molecules is known as polymer.
- What we have to know is between each of these small simple molecules the bond is a covalent bond now these polymer chains they could be linear they could be branched like this or they there could be a cross link network type of polymer.
- What is a monoma then?
- A simple molecule having two or more binding sites which can be linked together to form a giant molecule.
- When we say simple molecule, do all simple molecules classify as small modes?
- What is the structure of ethylene?
- CH2, double bond, CH2.
- What is the criteria?
- There should be two or more binding sites.
- Yes, we find that there are two binding sites across the double bond.
- The double bond can open and we get two carbons wherein the bonding can happen.
- Now here we have just one functional group, so therefore it cannot bind with several molecules.
- It can maximum m bind with one more ovastic acid and you just end up with a dimer.
- You definitely cannot get a polymer.
- No, there are no binding sites at all.
- What is ethylene glycol?
- Two OH groups, yes, if the binding can happen on both sides, therefore we could get a polymer from here.
- What is adipic acid?
- We have COOH, and here we have CH2 four times.
- And again we have COOH.
- So definitely there are two carboxylas carboxylic acid groups here, therefore it will act as a monument.
- Amino acid we have NH2 and this.
- CH2 could be N times, and then we have the C O H.
- Yes, here also we have two different binding functional groups.
- Therefore, this also will qualify as a monomer.
- What is styrene?
- CH to CH and there is a benzene ring here.
- So phenyl group, therefore, yes.
- Where does the binding take place?
- The moment you have unsaturation then we know that it is possible that you.
- This double bond can open up and you have two binding sites and therefore this also qualifies as a monomer.
- The next definition is of polymerization.
- The process by which monomers of low molecular weight are converted to polymers of high molecular weight.

- You can see there are double bonds, so many ethene molecules are there, and when they are polymerized, you get a chain like this and this is polyethylene.
- One chain will be referred to as a polymer molecule.

Example or application from the lecture

- Let us understand what is a polymer.
- Let us take up some examples and understand this.
- Let us take this example of ethene.

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EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 05:00–09:59

The total number of functional groups or bonding sites which are present in a monoma.

Key details

- Double bond CH2.
- Just now we saw it is CH2OH.
- Therefore, again it is bifunctional.
- Glycerol is CH2OH.
- CHOH and then again CH2OH.
- Therefore, we know that there are three functional groups through which binding can happen.
- Just now we saw it has an NH2 group and then CH two it could be N number and then we have COOH.
- Again there are two functional groups, so it is again bifunctional in nature.
- Degree of polymerization.
- How would you find out the degree of polyzation?
- If the molecular weight of the polymer and you also know the molecular weight of the monomer which you have used, we can calculate the degree of polymerization, molecular weight of polymer divided by molecular weight of the monoma.
- What would be the use of knowing the degree of polymerization?
- We can determine the length of the polymer chain and we can also find out what is the molecular weight of the polymer.
- That would be n because how did you define degree of polymerization?
- The total number of repeat units.
- One is the addition polymerization and the other is the condensation polymerization.
- Addition polymerization would be self addition of unsaturated monomers to each other takes place without the elimination of any byproducts.
- chain polymerization and examples would be PVC polystyrene teflon polyethene.
- C x is some other.
- Entity except so it could be H also, it is just some substitute substitution substituted group over here.

- Of course the bonding is happening on both the directions and therefore we have n to n of these units.
- Just by changing the substituent to Cl, we got polyvinyl chloride, which is totally different from polyethylene in properties.
- What do we get?
- We get is all four are fluorine atoms now.
- Instead of hydrogens, we are using fluorine atoms and what we end up will be Teflon, which is very strong polymer.

Example or application from the lecture

- Let us take them up one by one.
- Let us see the reaction here.
- For example, if we have hydrogen as the substituent, we end up with polyethylene.

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EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 10:00–14:59

and has exceptional properties and it is known as wonder plastic.

Key details

- Then what we get is polystyrene.
- Just by changing the hydrogen atom.
- You could change one atom, you could change all the four, you could change any number of hydrogen atoms, each time it is going to give you a polymer with totally different properties.
- We can have linear or branched polymers.
- Self addition takes place very rapidly, as I mentioned, it's also known as chain polymerization.
- There's no elimination of byproducts.
- Addition polymerization can take place by.
- It could happen by free radical mechanism.
- It is going to generate a free radical.
- in the monomer and that is going to help in the propagation of the reaction.
- That is not the only mechanism which can be followed for addition polymerization, it is also possible to do by cationic mechanism.
- A cation is generated just like in the free radical mechanism, free radical is generated.
- Here we are going to generate a cation.
- It is a Lewis acid.
- Now if the mechanism is anionic mechanism, then obviously now we understand anion has to be generated and therefore we are going to use some other agent now and that would be an organometallic compound.
- Only known organic organometallic compounds have Wrignax reagents.
- Or it could also happen through coordination mechanism.

- What is a coordinate bond where the covalent bond where the electrons are shared by one of the atoms.
- If the polymerization is happening through this mechanism, what would be used here?
- This is happening catalyst when we say it always happens on the surface.
- The reaction is going to happen on the surface.
- Polymer which is produced is going to be a very stereorcular polymer.
- For addition polymerization, we should know that molecular weight is an integral multiple of that of the monoma.
- The reason being that there is no elimination of any byproducts.
- We can have different types.
- It could be a homopolymer where you have just one type of monoma being used, or it could be a copolymer where you will have.
- Two or more different types of monomers being used.
- we could again get different types of co-polymers.
- Depending on how these monomers are arranged.
- It's not necessary that they have to be alternate.

Example or application from the lecture

- Now another example would be if we replace the X by a C6H5.
- So if we want to carry out the polymerization through free radical mechanism the initiator used has to be some free radical generator and that example would be benzoyl peroxide.
- That would be example would be ALCl3.

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EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 15:00–19:59

Let us look at them.

Key details

- When we say a single monomer unit, then it's called homopolymer.
- When we have two or more different monomer units, it's called a copolymer.
- it is called a copolymer.
- It is alternating.
- You can see one is say M1, the other is M two, and similarly they are alternating throughout the chain.
- That would be an alternating co-polymer.
- Or it could be random means.
- The copoly the two monomer units are just appearing randomly.
- There's no particular arrangement that would be a random copoly mark.
- Then we can also have a block copolymer where we have a block of one type of monomer, and then we have a block of another type of monoma.
- Then again, this block.
- Is coming up and then we have another block.

- Talking about condensation, polymerization, it would be intermolecular condensation through functional group of the monomers with continuous elimination of byproducts.
- And common examples would be nylons, polyester, and phenol formaldehyde resins all these are condensation polymers.
- Here we have the adipic acid and we have hexamethylene diamine.
- Hexamethlene diamine, when we use that, there is an elimination of the so here is sorry this is hexamethylene diamine and this is the adipic acid we see that COOH group here so OH from here and the hydrogen from the hexamethylene diamine is being used so hydrogen of the amine group.
- The linkage here is an amide linkage.
- CONH group is known as the MI group.
- Adipic acid and elimination of water molecule is going to give you this bonding between the two molecules it will take place and this rip unit is going to be repeated throughout the polymer chain.
- But what is the linkage we are getting here?
- We call it a polyester and the common name for using when you use terithalic acid and ethylene glycol is terylin.
- The as acids or alkali are being used as catalysts.
- In chain in chain polymerization, that is in addition polymerization we used initiators.
- We use the initiators like pre radicals, etcetera.
- Here it is cataly here we use a catalyst and this is also known as step polymerization.
- Chain polymerization is addition polymerization.
- Condensation polymerization is called step polymerization.
- Because the chain buildup happens stepwise, and there is continuous elimination of the byproducts.
- And one thing to note here is that linear branched or even cross-linked polymers are possible for condensation polymers.
- And of course, the molecular weight of the polymer is not an integral multiple.

Example or application from the lecture

- Now another example would be pterylene, where we have terithalic acid, again we have COOH here and ethylene glycol.

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EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 20:00–24:59

of the monoma because continuous elimination of these small molecules is happening.

Key details

- Differences between addition and condensation polymerization.
- Here there are small molecules are getting eliminated.
- Here initiators are required.
- Here we have catalyst and then linear or branch polymer are produced.

- Here we have cross linked also apart from linear and branched molecules.
- Weight is an integral multiple because there's no elimination of molecules, but small molecules here it is not an integral multiple of the monoma.
- And then monomers are unsaturated molecules are used as monomers and here they must have two active functional groups.
- All amorphous polybles when they are cooled below a certain temperature, they are found to become stiff, they become hard, brittle, and that state is known as the glassy state.
- But above the temperature, when it is heated, they become soft and flexible, and that is referred to as the rubbery state.
- This rubbery state when you heat it further then the.
- Polymer will start flowing, it gets into a viscous flow, and this temperature is known as the melting point of the polymer.
- One when it changes, you can see the graph here, one when it changes from the glass state.
- To the rubbery state.
- it when you heat it, when you're heating it inside, then it becomes into a viscous liquid and starts flowing, and when that change happens that is known as the melting point.
- Why is it changing like this?
- When we go to the very low temperature, I mean below the Tg, it is found that there is absolutely no segmental motion, neither the molecular motion.
- So it's like very stiff and therefore it behaves almost like a glass.
- Is that the glassy stint.
- You are giving energy to these molecules.
- But nevertheless, no molecular motion is here.
- So therefore only segmental motion.
- And this would be referred to as a rubbery state.
- But when you heat more, that means you will overcome all the forces of attraction between the polymer chains and the molecules now become absolutely free like this and therefore it change it becomes into a viscous liquid.
- If you want to fabricate a polymer or use it for a particular application, both TG and TM values are very important.
- We should know what are these values.
- So that means at room temperature we find that it is above the T G value and it is.
- Soft, whereas coming to polystyrene, we know polystyrene what it looks like, and there we see it is 100.
- So that means.
- At room temperature it is still in the glassy state.
- Where is the what would be the state at room temperature, and of course, when TG and TM you can fabricate and use this polymer for different applications.

Example or application from the lecture

- And that is given by Tg.

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Core explanation · 25:00–29:59

Properties of the polymer.

Key details

- In the next class, we are going to talk about the structure property relation between the in the in the polymers the relation between the structure and property of a polymer.

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EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

6 Structure-property relationship

Study focus: Structure-property relationship. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000078 · EV0000079

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module seven on functional materials, that is, polymers.

Key details

- We will talk about the structure property relationship.
- What do we understand by the term the structure property relationship?
- The properties of polymers are influenced by their structure.
- Polymers when it comes to properties are extremely versatile.
- Some can be hard, whereas others can be very soft and flexible.
- Some could be insoluble and resistant to heat, while others could be soluble and fusible.
- But here, normally the polymer will be in the form of this amorphous matrix.
- But in some polymers, we find that within this matrix also there could be some alignment like this where the polymer chains come very close to each other.
- Otherwise if those are missing then that is going to be amorphous in nature.
- Are the chains coming close to each other, is it possible for the chains to come close to each other and that would discur decide the crystallinity in a polymer.
- What could be the reason for a polymer to be crystalline?
- It depends on the structure to a great extent.
- CH2, CH2, CH2.
- combine each of these is CH2, CH2, CH2.
- Like this you have all these chains and since there is no obstruction for the chains to come close to each other we can say that this will be more crystalline as compared to a polyvenyl acetate.
- Polyvinyl acetate would be like this CH.

- O C O C H three.
- C H O C O.
- So you can imagine there will be another chain like this, and therefore these groups are referred to as bulky pendant groups.
- When you have such bulky pendant groups, obviously there will be an obstruction for the chains to come close to each other.
- Therefore such a polymer would definitely be less crystalline or more amorphous as compared to when there are no bulky pendant groups.
- Second reason is the presence of polar groups.
- Whenever there are polar groups, then it is found the polymer becomes more crystalline.
- Because whenever you have these polar groups in the.
- Polymer chain, we know that this could give rise to dipole interactions or hydrogen bonding.
- Polyvinyl fluoride, we know that the chain is like this, extending.
- And similarly you will have another chain here.
- So we find that because of the polar nature.
- We know that there is polarity here, H gets a small positive, and this bond is going to slightly shift here.
- So therefore, there is attraction between the between the chains.
- There is a dipole interaction which is happening between the chains, and therefore the chains can come closer whereas in polyethylene it is just as we saw just now CH2 and there is no such attraction except for the Van der Waals forces of attraction.
- Therefore we find that in the case of polyethylene it is.

Example or application from the lecture

- The properties we consider will consider will be crystallinity, tensile strength, elasticity, chemical resistance and plastic deformation.
- Let us take up the first property, crystallinity.
- For example, polyethylene.

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EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 05:00–09:59

much the crystallinity will be less as compared to polynomial fluoride.

Key details

- The crystallinity also depends on configuration.
- Ste is that is stereoregularity, crystallinity.
- When we say configuration, we are talking of stereoregularity, that is how are the substituents arranged in space.
- Here in syndioctic they are occupying alternate positions, whereas in A-tactic, they are just occupying any random position.
- The polymer is found to be most crystalline, whereas in A- tactic it is found to be least crystalline.

- What is the reason?
- Crystallin T is because you are the chains can come close to each other.
- We can see in isotactic they can come much closer to each other as compared to syntyotactic and a tactic.
- Coming to the next property, tensile strength.
- What would be tensile strength?
- The amount of stress a polymer can take before undergoing permanent deformation.
- By increasing the molecular weight we can understand that the tensile strength is going to increase.
- Like you can see in this graph, the tensile strength, when you increase the degree of polymerization or molecular weight, the tensile strength is increasing to a limit, and then after that it becomes almost constant.
- Increasing further would not make too much difference in the tensile stem.
- But one more property which we have to look at is the melt viscosity.
- As we increase the degree of polymerization of the molecular weight, the melt viscosity also goes on increasing.
- Remember that polymer.
- Something which we just test in the lab, they are for commercial purpose.
- Therefore, we need to make a compromise and therefore somewhere in this region where the tensile strength is also high.
- And the valve viscosity is not very high.
- This is the range which is used for fabrication of the polymers, I mean manufacture of the polymers, preparation of the polymers.
- PDC just now we know that there is a dipole attraction between the molecules, and when there is dipole attraction, obviously the chains will not slip each other slip against each other.
- Whereas like in the case of polyethylene, what did we notice in polyethylene?
- I just draw like this.
- This is the carbon chain which we have CH2, CH2, CH2.
- Because there's just Van der Waals forces.
- Tensile strength when you're applying a force on it, the chains will slip against each other.
- Whereas if you have some attraction, like in the case of nylon, between the chains we find that there is hydrogen bonding, there is a attraction that is.
- Hydrogen bonding between the chains.
- And of course, if we have cross linking, then.
- Definitely the inner polymer will also increase with the molecular weight and therefore there will be increase in the tensile strength.
- So therefore when we have cross linking like this, we find.
- Cross linking means there is covalent bond formation between the chains network three dimension network and therefore definitely it is going to this is going to result in increase in tensile strength.
- Cross linking also increases the tensile strength.

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EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 10:00–14:59

This is the next property we need to consider.

Key details

- The property of a polymer by which it deforms on applying force and regains its original shape when we when the force is released, and that is known as elasticity.
- There is a.
- it goes to the structure again.
- They have long coiled and entangled chains, and when you.
- Apply force, you are straightening them and they get oriented like this.
- So if they are long coiled chains like this entangled, but when you apply force they get stretched like this, therefore they become hard, and then when you release again they go back to the coil form.
- Now presence of polar groups of course is going to result in polymers which will be very stiff and inflexible because of the strong force of attraction which will exist between the chains.
- Now if nonpolar groups are there then it is just weak wanderwald forces and therefore they will be much more.
- We can say elastic in nature and presence of bulky aromatic and cyclic groups of is also going to result in decreasing the elasticity of a polymer.
- If we have an elastic material, can we increase its elasticity?
- That can be done with the help of cross linking as is done in the case of vulcanization.
- What is vulcanization?
- It is treated with sulfur so that the sulfur linkages will be.
- Obtained in between the polymer chains, and since these are covalent bonds, obviously it is going to give much more strength, and when we apply force the chains are not going to slip against each other.
- But if we introduce a small amount of cross linking using sulfur we find that is called balconized rubber and we find that these chains are not going to slip against each other and definitely the tensile strength can be very increased to a good extent.
- Of course we should not introduce too much of vulcanization because instead of becoming elastic it is going to.
- To become stiff because you have increased too much of force of attraction covalent, too many covalent bonds between the chains that is going to result in making it more stiff.
- We have elastic materials.
- But if the material is non-elastic, meaning it's very stiff, can we make it more elastic?
- Yes, that is done with the help of plasticizers.
- What are plasticises?
- They are small organic molecules which are made to enter between the chains.
- Now this can enter between the chains and when it enters between the chains it's understood that the distance between the polymer chains will increase and therefore it will make it less stiff and more flexible.
- The next property is chemical resistance.

- The resistant to resistance to swell, dissolve, and get degraded in the presence of a solvent or a chemical, and it will depend on the chemical nature of the monomeric units which are being used.
- And their molecular arrangement.
- If there are polar or non-polar groups.
- Because it can easily form hydrogen bonds with water.
- It's all like dissolves like.
- Instead, they will easily swell and sometimes dissolve in non-polar solvents such as petrol, benzene, carbon tetrachloride, etc.
- The second one could be.

Example or application from the lecture

- Suppose this is a polymer chain of natural rubber, what is done in vulcanization?
- These small molecules like for example triphenyl phosphate.
- For example, PVA polyvinyl alcohol will dissolve in water.

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Core explanation · 15:00–19:59

is the residual unsaturation.

Key details

- Whenever there is residual unsaturation, it leads to oxidative degradation.
- If there is a double bond in the polymer.
- See addition polymerization when we are doing monomers will always have double bonds.
- That is natural rubber, but when that is the monomer unit in natural rubber.
- But what do we notice here in the polymer itself there is still a double bond.
- When you have such a situation.
- Because of the nature of this double bond.
- In the polymer there is still a double bond, and we know that pi electrons can easily be take part in reaction.
- Therefore, such polymers will undergo oxidative degradation and.
- Because of this residual bond which is there, residual double bond which is there or unsaturation which is there, it gets easily degraded in air and UV light.
- And of course.
- How is dense is the packing?
- Because when there is dense packing in the molecule, then it will not allow the solvent to penetrate between the layers, as in the case of say Teflon.
- We know that Teflon, this is the.
- What happens is they are so close to one another.
- Because of the polarity.
- Each of these bonds is polar in nature.

- CF, CF, CF, all these bonds are polar in nature, and therefore the chains are going to be pulled very close to each other.
- So therefore, it will not allow any solvent.
- To get in between the chains.
- Therefore, we find that Teflon if water cannot enter, so it acts as non-stick.
- Polymer being more chemical resistant.
- That is plastic deformation.
- Some polymers have this property that when we apply heat they become soft and flexible, but on cooling they again return to their original shape.
- This is known as plasticity.
- A thermoplastic polymer would be one which softens on heating and becomes hard on cooling and it can be remoulded into new shapes and that is as the case is of polyethylene, polypropylene, etc.
- But what is a thermosetting polymer it becomes hard on.
- it initially it will be fusible, it becomes fusible, but the moment you cool it first time itself it is going to set into a hard mass and after that it cannot be changed anymore, it cannot be re molded anymore.
- Such polymers are known as thermosetting polymers and it they if at all you heat it then it is going to undergo.
- Total degradation that means it will not melt, rather it is going to degrade into some other compound.
- These are all thermo examples of thermosetting plastics.
- What is the reason?
- Why are they behaving like this?
- It has to do with the structure.
- There could be underwhelm forces, it could be dipole interaction or hydrogen band bonding between the chains.
- So when you are heating then you can easily overcome the force of attraction and therefore you see here when you have a thermoplastin, then when you heat these chains become loose, you have overcome the force of attraction and therefore they start to melt and the moment you reduce the heat or you c cool it, they will come back to the original shape.

Example or application from the lecture

- Like for example isoprene.
- And of course, molecular mass for a given polymer, as the molecular mass is increasing, we find that the swelling character decreases, and of course, if there is decross linking, if there is a high amount of cross-linking in the polymer, the solubility definitely is going to be less.

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EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 20:00–24:59

And in the case of thermosetting, we know that in a thermosetting it's a three-dimensional cross-linked structure which is there.

Key details

- And there are these covalent bonds which are existing between the chains.
- And now when you heat it, we cannot overcome this force.
- Of covalent bond because it's very high.
- Even if you do, then it is going to break into different types of molecules.
- Once these cross link structures are formed, we will not be.
- We are able to get them into.
- get them again remould them again and again.
- Here weak forces of attraction exist whereas here there are covalent bonds and we cannot when we heap them we cannot overcome these strong forces of attraction.
- That is a covalent making the covalent bond.
- What we understand is that the properties of the polymers are dictated by the structure of the polymer.
- So in the next class we are going to talk about some commercial polymers and we will take up butile rubber.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

7 Synthesis and properties of epoxy resin

Study focus: Synthesis and properties of epoxy resin. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000098 · EV0000099

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module 7 on functional materials on polymers, and today we are going to study about adhesives and specifically about epoxy resin.

Key details

- Only when it solidifies then the two surfaces get locked to each other.
- The examples would be phenol formaldehydrasen, urea formal dehydration, and epoxytes.
- How do these actually work?
- After that, the most important step is to add a curing agent.
- What does this curing agent do?
- It is going to trigger cross-linking into the adhesive.
- And because of cross-linking three-dimensional network which is formed, solidification happens, and what you get is a thermoset solid polymer, and that is how the two surfaces got bound together.

- Addition one is the low molecular weight polymer, which is soluble, and the other one, which we usually refer to as hardener, is actually known as a curing agent, and that is responsible for the solidification of the.
- Polymer and that happens due to three-dimensional cross-linking network getting form.
- There are certain materials for which you have to use a particular type of adhesive.
- For wood, it's usually urea formalhydrasin, but epoxy resin is a very good resin because.
- because it can be used to attach metal, wood, glass, concrete, leather, any surfaces, epoxy resin can act as a very good adhesive.
- What is an epoxy group?
- It is a three-membered ring.
- of an O atom bonded to carbon atoms.
- Three member rings are usually very reactive.
- Now, bisphenol A is the first monomer which is being used.
- When these react, what we get is the bisphenol A.
- The other monomer is epi chloroth.
- We add Cl2, what we get is the allyl chloride.
- That is now treated with HOCl.
- Dehydration is going to happen.
- It's sorry, it's Cl will be removed, and you get a three-membered ring over there.
- So this is known as Epichlorohydrine and you get NaCl because you are treating with it NaOH plus this HCl so you get the NaCl.

Example or application from the lecture

- Let us first define adhesives, yes, substances which are used to bind two or more materials firmly by surface attachment due to the penetration of the adhesive into the material and solidification which leads to interfacial locking.
- For example, for rubber we use phenol formaldehydrin.
- Let us see the synthesis.

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EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 05:00–09:59

And what do we notice here that there is an OH group over here and a Cl.

Key details

- These two molecules can undergo condensation, and this would happen in the presence of NaOH.
- HCl is getting eliminated.
- And what would happen as a result of this condensation?
- We can see here that there is an attachment between the bisphenol A and.
- And the epi chlorhydrate.
- This ring has opened up because if you look at this monoma.
- There are two hydroxyl groups on both sides, therefore it would classify as a monomalase.

- In the first class I mentioned, at least two functional groups should be there.
- But in this epiclorohydrate, we find there is just one CL which can take part in the condensation, whereas here we don't find anything, but yes.
- That is the reason why we have the ring here.
- Now this ring being very reactive, it is going to open up when this condensation will happen and therefore this ring is going to open up like this and we will find that we get the ch oh you can see here the ch oh is formed and now this ch2 is free to condense with the next bisphenol A ring.
- So here now this we have Cl condensation on one side and the other side it is CH2 where the ring is opened up and there's a free bond here that is going to react with the next bisphenol A.
- So this is the mechanism, and therefore we get this as the main repeat unit in the epoxy polymer.
- Epoxy resin is formed, an excess of epichlorohydrine is used.
- What would be the reason for that?
- The reason is that the N groups we want them to be epoxy groups.
- make sure that the that happens by using an excess of epi chlorohyain.
- When we use excess of epichlorohyden, because there is no more bisphenol, further reaction with bisphenole is not possible.
- If there was more bisphenol as we understood the mechanism just now, this ring would open.
- We'll get a CHOH and the CH2 combined would have combined with another bisphenole, but there is no bisphenol, so the ring remains intact, and in the end, we have ensured that both the side chains, the end of terminal chains, terminal groups are epoxy groups.
- That because we know that epoxy is a very reactive group, and when we add the curing agent, immediately there will be a.
- Three dimensional cross link structure which is formed, and we know that when between these two chains we have cross-link structure getting formed, the polymer is going to solidify.
- Use we use diamines, di carboxylic acid, or acid anhydrides.
- Because there are these reactive epoxy groups, immediately cross link structure will be formed, which will result in solidification, and hence the two surfaces are going to get bound.
- What are the properties of these epoxy resins?
- They have, of course, excellent addition, excellent resistance to water, acid, alkali, and other corrosive chemicals.
- And they also have very high mechanical strength.
- They absorb less moisture.

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EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 10:00–14:59

and they have very good insulation properties.

Key details

- They are also used to bind glass, metal, leather, wood, ceramic, etc.

- Of course, that is the reason why they come under the class of adhesives.
- Then when they are coated on the floor, they give and anti-skid layer.
- They are used for laminating materials and for crease and shrinkage resistance in fabric.
- So in the next class we are going to talk of another important class of polymers or rather polymer composites.
- So we will take up Kepler in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

8 Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 1)

Study focus: Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 1). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000110 · EV0000111

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module seven on functional materials on polymers, and in today's class we are going to study about polymer composites.

Key details

- We will look at Kevlar synthesis properties and applications and then we will study about reinforced Kevlar composites.
- In a composite we will have two or more constituent materials and their properties are not attainable by each of these components acting alone.
- It's only when they are mixed together and form a single material that is known as a composite.
- When one or more of these components are polymers, we call them as polymer composites.
- There is always a fiber and a matrix.
- The matrix would be.
- Polyester, epoxy, phenolic, silicone, melamine, or a vinyl derivative.
- And the fiber could be a glass fibre, boron filaments, carbon, graphite fibers, aromate, that is aromatic polyamide, or aluminium fibre.
- With these two components, the material which is made is known as the polymer composite.
- They are produced by suitably bonding the fibre material with the polymeric coalescent matrix and curing the same under pressure and heat.
- Because they have very high strength per unit weight.
- It is no it is an aromate fibre.

- What do we mean by aromate fibre?
- It is aromatic.
- It is because of the components which are being used, because of the monomers which are being used to make the polymer.
- Condensation of two monarchs that is one for phenyline diamine and one for benzene di carbone fluoride.
- thalic acid because these are in the para position the COOH COH groups are in the parapositions so we call it as steal acid and instead of the thalic acid we are using the acid chloride so this becomes steady thaloal chloride.
- That is because the reactivity will be much more enhanced when we use the teritaloid fluoride.
- It is 14 phenyline dye.
- Again, these two are in the para position.
- The amine groups are in the para position.
- polymerization which will take place, the condensate condensation reaction would.
- Result in elimination of an HCl molecule.
- So like this, these two monomers are because they have.
- Two functional groups each.
- So if this condensation happens, this side is still open for the next diamine to attach, and similarly this NH2 is open for the next.
- COCL group to attach.
- So therefore the polymerization continues this way, and what we get is amide linkages between these monomers and we get Keblar.
- How did we get it?
- It is a condensation polymer.
- Kevlar is known to be five times stronger than steel on an equal weight basis.

Example or application from the lecture

- Firstly, let us understand what is a composite.
- Let us study about Kevlar, which is a very important fibre.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 05:00–09:59

All of us know that Kebler is used to make bulletproof jackets.

Key details

- The first reason is that there is extensive intermolecular hydrogen bonding which is taking place between the Kevlar chains.
- chain and so on so many chains are there in Kebler now between these chains there is formation of hydrogen bond now we know that between the polymer units itself the when we join the units monomer units the leakage which we are getting is an amide linkage.
- Then again here we have CO NH.

- NH C O.
- So this is getting repeated, and here again in the same way in the next chain also we see there is C O N H N H C O C O N H N H C O.
- This is one thing we need to pay attention to that once we have.
- NH group here and CO here, the next one cannot be NH and CO, it has to be CO and NH because it is formed from the diamine, so NH and NH and then again we have CO that is the amide linkage here.
- And then so we see that there are amide linkages and but between the cablar chains now we have strong hydrogen bonding which is taking place.
- How is it happening between the carbonyl groups and the NH centers.
- that the hydrogen and the oxygen the hydrogen of the amine and the oxygen of the CO are coming very close, they are very in close proximity and that would result in formation of hydrogen bond.
- So there are so many hydrogen bonds possible because repeatedly every unit there is a CONH group and these N of one the hydrogen of one of the chains and the.
- O of the other chain they are coming in close proximity, and that will result in formation of hydrogen bond.
- Now because of this aromatic ring, this is giving a lot of firmness to Kepler.
- If it's an elephatic chain, we know it will be more flexible.
- Whereas now it is in the form of an aromatic ring.
- And that is giving firmness to Kevlar.
- And another reason for this is aromatic stacking.
- To the polymer, and therefore we find that keevlar is very strong.
- The answer is there are two reasons.
- One is the extensive hydrogen bonding.
- between chains of kevlar and additional strength is coming from aromatic stacking.
- What are the properties of Kevlar?
- We saw the structure and based on that we know that properties are going to depend on the structure.
- It has high strength to weight ratio and very high chemical inertness, low coefficient of thermal expansion.
- It also has flame resistance and high impact and cut resistance and textile processability.

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EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

Core explanation · 10:00–14:59

Based on all these properties now we can understand that there will be a lot of applications of Kevlar.

Key details

- Then gloves, sleeves, jackets, etcetera are made of Kevlar to protect users from cuts and heat.

- And then inner lining in bicycle tires, then lightweight boot hulls, aircraft panels, race cars, bridge structure.
- Earlier they used to use steel pipes.
- Now we know that when this is in happening offshore there will be in the ocean and therefore there'll be a lot of salinity and corrosion is very rampant over there.
- ropes in offshore drilling.
- What are they?
- Special cutting scissors and drilling machines are required because it's a very strong material.
- And one more problem here is that the fibre is hygroscopic.
- We have still not formed the composite because when we talk of a composite, we know there are two components.
- Obviously, we need a matrix for that and that matrix is usually an epoxy resin.
- And now the material will become so strong that it is going to now have many more applications.
- This matrix when you have and these fibers are embedded in it, then any force which is applied onto the material will be dissipated the impact is going to be taken by the metric so that makes the material even more strong.
- This is known as reinforced Kevlar composites.
- tennis and tennis and squash rackets.
- So now the applications have increased because the material is still more stronger as compared to Kepler.
- Kevlar on the next class we will talk of another polymer composite and that is carbon fiber.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

9 Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2))

Study focus: Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000122 · EV0000123

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module seven on functional materials, that is, polymers.

Key details

- Today we will talk about another polymer composite that is carbon fiber.

- We talk about the synthesis, properties and applications and finally we will look at the composite of carbon fiber which is known as carbon fiber reinforced polymer.
- What are carbon fibers?
- Basically, they are fibers which are made mostly of carbon atoms, and the diameter is just about 5 to 10 micrometers.
- When you compare this diameter with the human hair, as you can see in the picture here, the human hair is much thicker as compared to the carbon fiber.
- On polymerization is known as polyacrylonitrile.
- So now the next step, how we are going to use this polyacrylonitrile to get the carbon fiber.
- So here again the first step would be polymerization which I mentioned just now.
- So all these acrylonitrile molecules are going to be placed like this because it is renal cyanide.
- The polymerization will always take across take place across the double bond.
- So you can see the chain has got formed because of the opening of this double bond.
- So that is the first step which is.
- Now the next step is we need to cyclize it.
- Remember that we want a graphite like structure.
- So the next step would be cyclization that is done at a low temperature.
- Cyclization process there will be a bond which will be formed between this carbon and this nitrogen.
- Up and you get a bond getting formed between C and N, and here it will be left with double bond.
- That's exactly what you can see in this picture, and that will result in a cyclic molecules, and therefore that is called as the cyclization process.
- Polymer now has become cyclic is constituted of cyclic chains, or you can see here cyclic rings, and these are in the form of a chain.
- So now the next step would be.
- I want a double bond here.
- Because we want to go towards a graphite like structure.
- I want a double bond here.
- How do I get it?
- That is done by the oxidative treatment.
- This is the cyclization, and the third step would be oxidative treatment, which is done at a higher temperature.

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EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

Core explanation · 05:00–09:59

This is done at a higher temperature.

Key details

- Obviously, it is treated with oxygen and water molecule will be eliminated, and here basically we are removing hydrogen.
- Hydrogen that is unsaturation now came up.

- we are basically causing oxidation of this.
- this small this chain polymer over here.
- So now the next step is the most important step where we are going to go towards the graphite like structure.
- Of course the last step would be where you remove all other elements except carbon because we want a graphite like structure that is made of carbon.
- So graffiti is the final process and that is done at a very high temperature and in an inert atmosphere and inert atmosphere because you're treating it taking the fibre to such a high temperature, so it will it'll become inflammable, it may catch fire.
- We got some unsaturation in these positions by oxidative treatment and finally we heated it and we got the ribbon like structures joined together to give you this type of structure and finally when it is graffitized, that's called graphitization at a very high temperature.
- Then we get what we call as the carbon fiber.
- If you remember graphite is made up of it, it has a layer structure and.
- These chains between these layers actually there is a there is just Vanderwall force of attraction.
- But we do that carbon fiber is one of the strongest material.
- In graphite we have very regular pattern these lay of carbon they are just one below the other and because they're all sp2 hybridized so we have layers like this and these chains between the chains is just one wall force and the these chains can slip against each other.
- It is known as a turbostratic arrangement.
- What is that?
- These sheets of carbon fiber are half as a leaf folded or crumpled together.
- They are all folded and crumpled so as a result that slipping of chains is not possible and this is known as the turbostratic arrangement and that gives the carbon fiber a good strength.
- Say that it is also biocompatible.
- It is a fiber which is biocompatible and it has a low coefficient of thermal expansion.
- Of course, the most used property is high strength to weight ratio.

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EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 10:00–14:59

What would be the applications?

Key details

- It's very clear such a strong material we have.
- It is used in aerospace, road and marine transport, it is used in sporting goods and medical applications in prostesis surgery.
- Ligamentary repair of those.
- Because it is biocompatible.
- Then we also have automobile goods, novel tooling, casings and bases for electronic equipment, EMI and RF shielding brushes.

- But what we have learnt now is about the advantages.
- Of course, it is very expensive due to the large ex energy expenditure.
- Which are going to use carbon fiber, especially where a lot of strength is required.
- And one more disadvantage would be volatile byproducts are formed.
- We know that when we make a composite, there has to be a fiber that is the carbon fiber, and the matrix is the epoxy resin.
- That is how it is getting the strength.
- Of course, we know light weight and high strength to weight ratio, low coefficient of thermal expansion.
- But one new property which has come here and very important is the high electrical conductivity.
- Because we know it has a graphite like structure and we know that graphite.
- is conducting in nature.
- So because of that we find that it has high electrical conductivity.
- So in the next class now we are going to talk about a new class of polymers which are called conducting.
- Polymers so far polymers were always thought of as insulators, but yes, there is a class of polymers which are which can actually conduct as good as that of the metal, and these are known as conducting polymers.

Example or application from the lecture

- Like for example, H Cn could be formed when we are making carbon fiber from polyacrylonitrile.

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EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

10 Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same

Study focus: Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000134 · EV0000135

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module seven on functional materials that is polymers and this will be the last class of this module.

Key details

- In this class we are going to talk about molecular weight of polymers.
- That is number average molecular weight, weight average molecular weight, and viscosity average molecular weight.
- We know that its molecular weight is eighteen, and irrespective of the source from where we are going to take the water, the molecular weight is going to still be eighteen, and each of the molecule is going to have a molecular weight of 18, but the case of polymers is different because in a sample the molecular weight of the polymers, the chain length of the polymers is different.
- Each unit may have a different.
- Therefore, whenever we talk about molecular weight of a polymer, we talk about an average molecular weight.
- So this molecular weight now we know is a very important characteristic of a polymer sample because several properties of polymers are found to depend on the molecular weight.
- The entire range of mechanical properties which are very useful to determine the application of a polymer like stress strain, impact, fracture, fatigue, creep, stress relaxation, and cracking, they are all influenced by the average molecular weight and also the molecular weight distribution.
- There could be number average molecular weight, we can also have a weight average molecular weight and also viscosity average molecular weight.
- Number average molecular weight is denoted by Mn bar.
- The bar over M stands for the average.
- with the total number of polymer units present.
- Then the total weight would be obviously n1 multiplied by m1.
- Summation of NIMI divided by summation of NI would give me the number average molecular weight.
- Of course the summation is over I and what is I?
- It is the particular species we are talking about.
- So I will be 1 for the first species, so N1, M1.
- and divided by N1.
- And similarly, N2, M2.
- And then divide by the total number of units that is n1 plus n two plus n three and so on so forth, what we saw here.
- This is just writing it in a short form.
- Depend only on the number of the species and not on the size or the nature of the species.
- And such colligative properties you are already aware, like osmotic pressure, elevation and boiling point, depression and freezing point, or even relative lowering of vapor pressure.
- So all these could be used to determine the number average molecular weight because polligative properties depend only on number and when we are talking of number average molecular weight we are only concerned about the number of polymer units.

Example or application from the lecture

- When we talk of molecular weight of any compound, for example, water.
- Let us look at each one of them.
- Suppose N1, N2, N3 are the number of polymer units having molecular weights of N1, M2, N3.

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EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

Core explanation · 05:00–09:59

use the mass we are going to consider the mass of each of these species and not just the number.

Key details

- Why many times the number average molecular weight does not give you a clear picture of the average molecular weight.
- We want to we understand that the higher masses would contribute more to the molecular weight than the lower mass.
- In number average molecular weight here we are talking of the mass.
- So therefore mass average molecular weight on weight average molecular weight will be.
- M1 multiplied by capital M1 plus M2 multiplied by capital M2 and so on divided by the masses M1 plus M2 plus so on.
- Again, they are masses of the polymer units having molecular weights M1, M2, M3 respectively.
- M1 is replaced by N1 M1.
- So therefore it becomes N1 M1 squared.
- And in the denominator, we had M1 plus M2 plus M3.
- plus n two n two and so on.
- I stands for the that particular species.
- Obviously some experiment which depends on the size like how we chose polligative properties for.
- Number average molecular weight because colligative properties depend only on number.
- Here we are going to look for an experiment which is going to be dependent on the size, and these would be sedimentation velocity or light scattering.
- Sedimentation velocity will depend on the weight of the units and light scattering would depend on the size of the each of these units and therefore these two experiments could be used to determine weight average molecular weight.
- is known as the polydispersity index.
- This is known as a poly dispersity index.
- What does this tell us?
- It is a measure of the broadness of a molecular weight distribution of a polymer, that is, the larger the polydispersity index, the broader is the molecular weight distribution.
- Now if you had a mono dispersed polymer that means only one type of polymer unit.
- From the formula we can understand that weight average will become equal to the number average.
- How much is the weight average more than the number average?

- That is what we are talking about, the ratio.
- Weight average by the number average, how much is it different from one?
- That will give us the polydispersity index, meaning that if it was equal.
- If it all the polymer chains were of equivalent or very close to equal length, then we would have had PDI that is a polydispersity index is equal to one.

Example or application from the lecture

- There we consider number.

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EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

Core explanation · 10:00–14:59

But if they are different, then the ratio m w bar by mn bar, that is a poly dispersed index would be greater than one.

Key details

- Now looking at this picture here we see that these polymer chains there's very narrow dispersity, so therefore we can that is how we when we calculate the poly dispersity index, this will come very close to one, whereas here we see the polymer chains are of different lengths.
- This is known as the mark hoving equation.
- It is the same which is used in the Markov which comes in the Markoving.
- equation and this is how we determine using viscosity experimentally we can.
- Determine the viscosity average molecular weight.
- And this value of A is going to be dependent on the polymer solvent system and also the temperature.
- And usually the value lies.
- between 0.5 and 0.9.
- Because we know that larger molecules will contribute more to viscosity.
- And we found that weight average is always higher than the number average molecular weight.
- This peak here, when we get a bell shaped curve like this, the peak will represent the number average molecular weight, and since this is the molecular weight scale, we find weight average is always more than the number average molecular weight.
- B so that is the reason that reason I explained just now that because viscosity.
- The larger molecules contribute more to the viscosity of the system and therefore viscosity average molecular weight will be closer to the weight average molecular weight.
- number average molecular weight.
- So with that we come to the end of module seven and in the next class learn about organic light emitting diodes.

Example or application from the lecture

- the molecular weight of each of the species and the number of polymer units of that particular species, then the equation is given this.
- Viscosity average molecular weight is given by summation of NIMI one plus A divided by summation of NIMI and the whole thing is raised to the power one by A.

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EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

11 Conducting polymers - examples, types of doping

Study focus: Conducting polymers - examples, types of doping. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000146 · EV0000147

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module seven on functional materials that is polymers and today's class we are going to talk about conducting polymers, a brief introduction, the types of dopamine, the applications of conducting polymers.

Key details

- Polymers are generally known for their insulating property because the organic molecules of which they are composed have no free electrons to carry current.
- An organic polymer with highly delocalized pi electron system and having an electrical conductance of the order of an electrical conductor.
- Such a polymer was referred to as a conducting polymer.
- We know that copper has very copper metal has very high conductivity of the order of 10 raised to the power of eight sements per meter, whereas Teflon has conductance of 10 to the power of minus 8 siemens per meter.
- And if you have doped polyaniline, it is found to have 10 to the power of 5 sements per meter.
- These are also known as synthetic metals.
- Heger, McDermott, and Shirakawa won the Nobel Prize for Chemistry for the du discovery of conducting polymers.
- The answer is definitely no.
- poly thiophene, polyaniline, and here we can see polyphenylin vinyline.
- what we can see they have a linear structure.
- All those polymers which we listed just now they have a linear structure and here's I've taken just two examples.
- Which should be there in their structure, and that is extensive conjugation in its polymer backbone.
- What would that mean?

- That they should have alternate.
- single and double bond.

Example or application from the lecture

- So let us talk about the development of conducting polymers because just recently in the year 2000.
- And let us have a look at some examples of such polymers.
- For example, polyacetylene, then we have polyphenylin, then this is polypyro.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

Core explanation · 05:00–09:59

When what is doping what are we doing here?

Key details

- We are introducing a charged chemical species, we find that the conduction increases manifold.
- There could be oxidative doping, there could be reductive doping, or there could also be proton doping.
- The pi backbone of the polymer is partially oxidized with a suitable oxidizing agent.
- It is going to take away electron from the polymer backbone.
- And now we understand why we chose a.
- Pi system where because we know that pi electrons they can easily be removed from the they are more mobile or as we say as compared to the sigma bar sigma electrons, sigma bond the electrons which are forming the sigma bond.
- See these plants are much more free to move.
- So therefore they can be easily removed from the system.
- That's why we chose the pi electron system here.
- This creates a positively charged site on the polymer backbone and now these become the current carriers for conduction.
- They are going to remove an electron from the pi system, hence we get a positive charge over here.
- In order to compensate for the positive charge, we know that you have learned during different mechanisms that these electrons will move to the next bond, these will move to the next bond, hence the electrons are moving in one direction and the hole is moving a positive charge or the whole is moving in another direction just like it happens in.
- P type doping and what is moving along the chain when a potential is applied is the electron or this positive charge, they are moving in opposite direction, and that is how the polymer becomes conducting when we do oxidative doping.
- The mechanism how it is happening.
- So when a potential is applied, so again because these electrons can be delocalized, so these electrons now move in the opposite direction, and therefore this negative charge moves in one direction, and again the hole is going to move in the direction.

- the opposite direction.
- So because of this pi electron system, the alternate extensive conjugation which we have in this polymer backbone, this type of movement is possible.
- The third type of doping is known as protonic acid doping.
- And what is here the current carrying charge species that is the positive and the negative species are again created and that is with the help of a protonic acid.
- The current carrying charge species that the that is the positive and the negative species are created by protonation of imine nitrogen.

Example or application from the lecture

- And that is how now we have negatively charged sites which are moving and hence we call it as N type of doping and this can be done with reducing agents like for example sodium naphtalide in tetrahydrofural.
- Now example best example would be doping of polyaniline and in polyaniline.
- Let us just have a look at what the structures look like.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 10:00–14:59

Partial oxidation is done with the help of ammonium peroxy disulfate.

Key details

- So you will have alternate this will be alternated along the entire polyanyline change chain and now what we are going to do the next step is a protonation that is done with H Cl that is one molar H Cl.
- H plus i that's why it's called protonic acid doping.
- This gets the plus charge and now it is balanced with the help of Cl minus.
- Cl minus so this is now becoming a plus n h plus n h plus and now this is balanced with the help of Cl minus and here also Cl minus because we added H Cl.
- So this is how it got doped, and now because of the delocalized pi electron which are available here, we can have these charges moving along the along the chain.
- Emeraldine salt and this is green in color and this structure is what is conducting polyanine so you can write here conducting polyanum.
- So as such, in this case we don't find it to be conducting the first structure.
- When it undergoes partial oxidation, what we get is known as the amoraldine base.
- This is also not conducting.
- It's only after protonation when the MRL wing salt is formed, then.
- This is this is found to be green in color, and this is the structure of conducting polyanuline.
- And how did we make it conducting?
- We know now that polymer are conducting in nature some of these polymers and they can be made conducting by doping.
- But when we talk of electrodes, we are talking of electronic conduction.

- Now what we saw in conducting polymers is electronic conduction because the electron could move through the polymer chain.
- So therefore they will be used as electrode material and because they are lightweight we expect a very high power to weight ratio.
- Then they can be used as conductive tracts on PCBs normally we see copper being printed there.
- Then they can be used as sensors.
- Like humidity sensor, gas sensor, radiation sensor, biosensors, etc.
- Then electrochromic display window, specially polyaniline, is known to have different colours in different atmospheres.
- Now the color of each of these is different and therefore it can be used in electrochromic display windows to tell us what atmosphere is there inside the system which is which is being studied.
- Then in information storage devices as film membranes for gas separations in light emitting diodes and they are important materials in microelectronics applications, then electro catalysis fuel cell electrodes and reinforced composites etc.
- And now in the next class we are talking about the molecular weight of polymers.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

12 Organic light-emitting diodes (OLEDs): Construction, working principle, and light emission mechanism

Study focus: Organic light-emitting diodes (OLEDs): Construction, working principle, and light emission mechanism. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000158 · EV0000159

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will study module 7a today about organic light emitting diodes.

Key details

- This is the only class in this topic, and we are going to study about the construction, working, the types of OLEDs, the advantages, disadvantages, and applications.
- The most common material is gallium arsenide, which is used to make these LEDs.
- Further doping is done and we have different types of colored LEDs.
- And they recombine in the depletion layer.
- When they recombine, they give out a photon and that is what we see as light.
- They use organic molecules and hence the name organic light emitting diodes.

- The phenomena which is happening in OLEDs is called electroluminescence.
- Therefore, OLEDs have an emissive electroluminescent layer, which is a film of an organic compound that emits light in response to an electric current.
- Hence they are also known as organic electroluminescent diodes.
- The first OLED device was made by Eastman Kodak in the nineteen eighties.
- So we know Kodak Labs, it was first made there and it's an emerging technology for displays.
- All of us have seen.
- OLED TVs, so much advertisement for OLED mobile displays.
- Because these displays can be flexible, they can be very thin and of light weight.
- Moreover, the power consumption is less as compared to the conventional LEDs and the L C Ds.
- Apart from that, they offer a high contrast, brighter displays with wider viewing angle.
- And here you can actually roll it.
- First of all, there is a substrate which is made of glass and that is basically for support.
- On the substrate, we can see the anode.
- This is usually transparent and made of indium tin oxide it is ITO and then we have two organic layers here as you can see this orange layer over here is known as the conductive layer and it is going to transport holes from the anode and this is usually made of polyaniline and also there are other material organic molecules which are used to make this conductive layer.
- And then we also have the next layer which is again an organic mole made of organic molecules and this is known as the emissive layer.
- And here we have these this is very important because this is where the light emission happens.

Example or application from the lecture

- First, let us understand the working of light emitting diodes, because the same concept can be extended to organic light emitting diodes.
- Let us understand the construction.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 05:00–09:59

and light is emitted.

Key details

- And the cathode of course is used to inject electrons and it is usually made with metals like magnesium which have a low work function.
- A voltage is applied across the anode and cathode of the OLED.
- We can see that the cathode will be connected to the negative terminal, hence there are.
- Electrons coming from the cathode.

- The green layer you can see here, these are moving to the emissive layers from emissive layer from the cathode.
- And at the same time, electrons.
- which have moved out of this orange layer over here which is the conductive layer.
- So therefore what happens?
- Holes have now moved into the conductive layer.
- You can see here closer to the emissive layer and therefore light is being emitted from the emissive layer.
- And the intensity of the light, that is the brightness of the light, is going to depend on the current density.
- These two are going the holes in the electrons are going to combine and this combination happens closer to the emissive layer and that is how whenever there's a combination of a electron and a hole then there will be emission of light.
- And the holes are at a lower energy level.
- We have learned this in right from spectros molecular spectroscopy when we were studying.
- We have passive matrix OLED, which is known as the emolet, and we have the active matrix OLED, that is the emolet.
- Then we have transparent OLEDs, we have foldable O LED, we have top emitting OLED and white O LED.
- The first one is a passive matrix O LED, and here we have strips of these cathode.
- and anode and the organic layers.
- The anode strips are arranged perpendicular to the cathode strips and the intersection of the cathode and anode they form the pixels from where the light is emitted.

Example or application from the lecture

- One example of a polymer which is used.
- Whenever there is an electron transition from a higher level to the lower level, the energy is given out in the form of a radiation, and here it is in the visible region.
- Let us look at each one of them carefully what are these different types of O LEDs which we are referring to.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

Core explanation · 10:00–14:59

There has to be an external circuitry here which applies current to selected strips of anode and cathode, thereby determining which pixel should be turned on and which pixel should be turned off.

Key details

- So because of this external circuitry here, the power consumption is more compared to the other OLEDs, and therefore these are mostly used for small screens.
- Coming to active matrix OLEDs which are more popular and many of the mobile displays now they use this amulet.

- They have a full layer of cathode, organic molecules, and anode.
- Here the anode layer has a thin film transistor TFT array at the back plate.
- There is a back plate of TFT and that forms a matrix.
- Now amulets therefore consume less power than the P amulets because the TFT array requires less power as compared to the external circuitry and therefore they are very efficient for large displays.
- like we have electronic signages or we have these big TVs, large screen TVs.
- They consist of a transparent substrate, cathode and anode.
- And a transparent OLED display can be either active matrix type or a passive matrix type.
- And it is used in transparent projector screens and glasses.
- The next one is a top editing OLED.
- They are best suited for active metrics design and they are used for making small card displays.
- Then foldable O LEDs.
- They have the substrate made of very flexible metal foils or plastics, as you can see here, and they are very lightweight, ultra thin and because they are foldable they can reduce the breakage of displays.
- Personal digital assistance.
- They give you true color quality like of incandescent lighting, and therefore they can be used for large and efficient lighting systems.
- The organic lay of an OLED are thinner, lighter, and more flexible than the crystalline layers of LED or L CD.
- We know that in the market we have L C D or.
- TVs and then came the LED TVs.
- And now there's a lot of talk about O LED screens.
- They are brighter than LEDs because the organic layer of an O LED are much thinner than the corresponding inorganic crystal layers which are used in LED.
- And so the conductive and the emissing emissive lay of an OLED can be multi-layered.
- And of course they consume much less power than the L C D's.
- and then easier to produce and can be made in large sizes and they have large fields of view.
- In an L C D we know that if we go at an angle we will not be able to see the picture.
- That is not the case with O L E D.
- Up to 170 degrees the viewing angle has now become very large and there's a very high color contrast and faster response time.
- They have shorter lifespan and the manufacturing process is expensive.
- And one of the most important problem we face with OLEDs is water can easily damage them.

Example or application from the lecture

- Let us see what are the advantages of OLEDs.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000166 · EV0000167 · EV0000168 · +1 source units in pack.json

Core explanation · 15:00–19:59

What are the applications?

Key details

- They are used in mobile phone displays.
- And then we have digital cameras where they can be used.
- As we said, white bullets, they can be used for lighting.
- Then we have audio video players and then we have OLED TVs and then also different.
- Other type of devices like wristwatches, so displays in wristwatches, etc.
- And if they are foldable, you can put it right over your wrist.
- And in the next class we will be talking about green chemistry, which is the last class of unit five.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

13 Green chemistry: Introduction, twelve principles of green chemistry with examples

Study focus: Green chemistry: Introduction, twelve principles of green chemistry with examples. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000174 · EV0000175

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we will look at module eight and that is about green chemistry and this is the only class in this module.

Key details

- We will study the definition and the twelve principles of green chemistry.
- Green chemistry is defined as the utilization of a set of principles that reduces or eliminates the use of use or generation of hazardous substances in the design, manufacture, and application of products.
- We all are aware of environmental chemistry where we talk about removing eliminating pollution removing pollution from the environment.
- But here green chemistry talks about.
- Removing pollution, yes, but not generating it in the first place itself.
- Anistos and John C.
- The what is the focus of green chemistry?
- Then we have to find creative alternative routes to minimize impact on the environment.

- And it also, of course, provides sustainable development, sustainable businesses, and sustainable living practices.
- Prevent waste, atom economy, safe synthesis, safe products, design benign chemicals, design for energy efficiency, use of renewable feed stocks, reduce derivatives, catalysis,.
- Designed for degradation, pollution prevention, and accident prevention.
- Of course the first thing we have to talk about when we talk about pollution is that not to generate waste in the first place because once the waste is produced that it involves an extra cost and time to treat and safely dispose it.
- And therefore the production cost also increases.
- The second principle is atom economy.
- In fact, in such a way the design should be of any reaction should be such that all the materials, in fact all the atoms which are present in the reactants should be incorporated into the products.
- That is atom economy.
- The emphasis was more on yield.
- Yes, here also we want a good yield, but at the same time it is not just the yield we are interested in.
- We want that all the atoms so that there is no extra atoms which are not being used or getting wasted.
- Some of the atoms are getting substituted by another atom.

Example or application from the lecture

- Like for example there's a substitution reaction.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 05:00–09:59

Whereas if there was an elimination reaction, definitely some small molecules are getting eliminated and or even larger molecules sometimes and then what happens to your atom economy?

Key details

- It has become low.
- If you have an addition reaction, we know that.
- Talking of green synthesis, then of course we understand atom economy must be maximum, and that is 100%.
- Safer synthesis is the third principle.
- As far as possible, we should use only substances which are innocuous and necessary safety precautions have to be taken while processing to protect human health and the environment.
- But to prepare the isocyanate we had to use phosgene and we know that phosgene is a very toxic gas, COCl2.
- It's a toxic gas.
- So the alternative now has been found and the green synthesis of polyurethane eliminates the use of first gene and instead CO2 is being used.

- It functions as efficiency and as efficiently, but the toxicity is reduced.
- or in pharmaceuticals this plays a very important role because many of these drugs which are being used for as medicine we know that there is lot of side effects.
- Then design benign chemicals.
- And if it's.
- All we have to use, then they should be harmless, innocuous chemicals which are harmless.
- We can use alternative solvents, find alternative solvents greener in nature.
- So many of the researchers are using these ionic liquids now organic solvents are usually avoided because they are volatile in nature and they can affect the health as well as the environment but when you have ionic liquids now these have the properties to dissolve.
- At the same time they will not be volatile.
- We have pyridine which is like this.
- with three bonds and there is a plus because we have attach an alkyl group here.

Example or application from the lecture

- Suppose there's a double bond and the reactants are getting incorporated into it, yes then we get hundred percent atom economy, and even if there is a rearrangement reaction again it is just rearrangement of the atoms which is taking place again we have a hundred percent atom economy.
- For example, polyurethane is a very popular polymer and it was manufactured conventionally using isocyanate and alcohol.
- One good example would be N alkyl peridinium cation with BF4 minus.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

Core explanation · 10:00–14:59

This alkyl group could be ethyl, methyl, and so on.

Key details

- Positive is now because of the N having an extra bond over here, this becomes positive and this is neutralized with the B at four minus.
- At any cost we have to avoid harmful solvents like chloroform and peritene, etc., which are known to cause health hazards, and some of them could be carcinogenic to human beings and animals and also cause severe damage to the environment.
- The next principle says design for energy efficiency.
- Say the reaction could be occurring at milder conditions or it requires less time to complete.
- For this purpose, it is worthwhile to utili something like biocatalyst or homogeneous or heterogeneous catalyst, which will reduce the energy of activation for the reactions and hence we will consume less energy.
- And modern techniques such as solvent free super.
- And more importantly these days microwave irradium and ultrasound.
- Biodiesel when it is made in the lab, the normal way, it takes nothing less than say one hour one and a half hours.

- But the same thing when we do in the microwave when we use microwave radiation to have that reaction or to perform the reaction when then that reaction takes place at a very fast rate and.
- And within 15 minutes we can have the same reaction happening which was taking one and a half hours.
- less energy intensive and saving more energy.
- Use of renewable feedstock.
- I think all of us are aware of this.
- Raw material or feedstock used as starting material for any reaction should be a renewable so it's should come from a renewable source rather than one which is depleting.
- Wherever it is technically as well as economically possible and for renewable feedstock the most common thing being looked at is biomass and agricultural waste which are strongly recommended rather than using the non-reduable sources like coal and petroleum based raw materials which we know are depleting very fast.
- This is very important because many times when reactions are carried out, we make derivatives.
- So temporary modification should be avoided whenever possible because what happens because of that?
- Going to generate waste, and at the same time, we know whenever we have these extra byproducts getting form the atom efficiency is reduced as the protective group is not getting.
- Incorporated in the final product.
- like this and I want to nitrate it.
- The moment I do nitration I will get ortho as well as the para substituted but what I want is just the ortho or the para.
- I usually take aniline and I'm going to treat it with CH3CO C L, so thereby making it into NH.
- So because of this, what happened now?
- We are blocking, we are protecting the NH2 group or making it less reactive.
- And now when we do nitration, what we observe is that either we get the para.

Example or application from the lecture

- For example, biodiesel.
- For example, we could be using some group to block or as a protection and then again deprotection.
- Let us take an example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

Core explanation · 15:00–19:59

Para product that is what we wanted.

Key details

- But then we wanted an ananine group, I mean NH2 group over here.

- in the paraposition and then NH2 nitroaniline that is para nitroaniline and ortho nitro aline aniline and not the disubstituted aniline.
- But what happened because of this you carried out so many reactions first of all and then you blocked this position and finally you have to unblock it like protection and deprotection now because of that a lot of wastage happens so we have to look for alternatives wherein directly aniline can be converted into orthodonitromine or paranitry.
- Anuline as we desire.
- of chemicals and also the time consumed in the reaction.
- Now another way of doing carrying out reactions is by catalysis and this is definitely a greener method.
- So we have to use catalytic reactions because catalytic.
- This we know are going to speed up the reaction, they can be recycled and are highly selective in nature.
- we can get very high yield and also not only high yield but we'll get very pure products.
- So therefore wherever it is possible we should look for reactions which can be carried out with the help of a catalyst.
- B of greener options because when we use tychometric reactions, they are highly energy intensive and they also require excessive reagents.
- The next principles talk about design for degradation.
- We all understand that insecticides and pesticides they will remain in and they are required for the plants but nevertheless they remain in the soil and gets absorbed into the earth and from the water there when it comes into the water bod and finally it is going to come back to us and affect us in a bad way.
- They're going to cause harm.
- So therefore, insecticides, pesticides, polymers, etc.
- We have to look for degradable alternatives for this so that they do not persist in the environment and are known to cause these generally will persist in the environment and they are known to cause bioaccumulations.
- Therefore, we have to find alternatives which are degradable so that they do not persist in the environment.
- Then pollution prevention.
- Here we are talking of.
- Real time analysis.
- Of course, all these points which we have mentioned so far about are about pollution prevention, but here we are talking of real-time analysis that is quality control.
- The definition of hazardous substances.
- This is the last principle, that is the twelfth principle.
- The chemicals chosen for the processes should be such that.

Example or application from the lecture

- This is an example where we talk about we have to reduce derivatives so that we can bring down the wastage.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

Core explanation · 20:00–24:59

We can minimize accidents in the form of fires, explosions, and toxic releases to the environment.

Key details

- Also, manufacturing plants should be well equipped with safety mechanisms.
- All of us are aware of the mobile gas tragedy.
- It is one of the worst industrial disasters and after so many years people are still suffering because of the leakage of a toxic gas into the environment.
- It is a set of principles which emphasizes on environmentally benign chemical synthesis with a view to devise pathways for the prevention of pollution.
- and then bringing a paradigm shift in chemical processing to achieve sustainable development.
- It is a concept which we should always keep in mind whichever field we are working in, that we have to save the environment, we have to keep it clean and safe.
- That brings us to the end of module eight and we have completed unit five.
- It also brings us to the end of this course on engineering chemistry.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

14 Source slide deck: Commercial polymers synthesis and properties of Plexiglass

Study focus: Source slide deck: Commercial polymers synthesis and properties of Plexiglass. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000401 · EV0000402

SHORT-NOTE RECAP

Study notes · Source slide deck: Commercial polymers synthesis and properties of Plexiglass

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING CHEMISTRY (UE26CY151 A,B)

- Department of Science and Humanities

Slide 2 · ENGINEERING CHEMISTRY

- Commercial polymers: synthesis and properties of Plexiglass

Slide 3 · ENGINEERING CHEMISTRY

- Polymethyl Methacrylate (PMMA) or Plexi Glass
- Plexi glass is obtained by radical polymerization in bulk, emulsion or suspension from the monomer methyl methacrylate (MMA)
- Hydrogen peroxide is used as initiator

Slide 4 · ENGINEERING CHEMISTRY

- It's Tg 1050 C, colourless, transparent hard solid at room temperature.
- Resistant to many chemicals but soluble in organic solvent like acetone and chlorinated hydrocarbons and esters
- Higher the optical clarity (100% Transparent in visible light radiation)

Slide 5 · ENGINEERING CHEMISTRY

- Due to its optical clarity it is used to make aircraft windows, signal boards etc
- For Making artificial teeth and contact lens
- Good substitute for glass but has poor scratch resistance.

Slide 6 · Department of Science and Humanities

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000401 · EV0000402 · EV0000403 · +15 source units in pack.json

15 Source slide deck: Experiential learning Based on nanomaterials

Study focus: Source slide deck: Experiential learning Based on nanomaterials. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000692 · EV0000693

SHORT-NOTE RECAP

Study notes · Source slide deck: Experiential learning Based on nanomaterials

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING CHEMISTRY (UE26CY151 A,B)

- Department of Science and Humanities

Slide 2 · Experiential Learning

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · Department of Science and Humanities

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000692 · EV0000693 · EV0000694 · +6 source units in pack.json

16 Historical question-source pages · unit mapping candidates

Study focus: Historical question-source pages · unit mapping candidates. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000701 · EV0000702

SHORT-NOTE RECAP

Study notes · Historical question-source pages · unit mapping candidates

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Archive page 46 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.6667; source text/OCR may be partial. OCR/sparse page: visual confirmation is required. Overview of Hydrogen Storage Methods Form Advantages Disadvantages O Gaseous Form - Less infrastructure required - Needs extra energy for compression - More cost-effective - Occupies a large space - Can be compressed to high - Requires leak-proof tanks pressure Liquid Form - Exists at a very low temperature - Reinforced and insulated storage tank required (20K / -253°C) - Energy-intensive cooling and compression (~30% loss) - Stored in cryogenic tanks - Low volume Solid Form - Hydrogen can be released on - Currently under intensive Research & Development demand (e.g., by heat) (R&D) - Ease of handling - Enhanced safety s s s Key challenges of the materials and their solutions 1. Weak Binding Forces: Hydrogen storage materials have high specific surface areas but weak binding forces (e.g., Van der Waals forces). 2. Metal Decorations: Metal decorations are effective in increasing the bonding energy of hydrogen on sorption-based materials. 3. Hydrogen

Spillover: e Hydrogen storage in carbon materials is enhanced by decorating these materials with transition metals. ¢ Definition: Hydrogen spillover refers to the migration of hydrogen atoms from the metal surface to anon-metal support or absorbent.

Archive page 1 from 10__1vFS6E1bKV.pdf; unit match candidate score 0.9167; source text/OCR may be partial. (4.0 Marks) (6.0 Marks) (4.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - JULY - 2023 UE22CY151B - Engineering Chemistry Total Marks : 100.0 1.a. State Frank-Condon principle. Draw the three possibilities of vibrational - electronic transitions energy level diagram and comment on the spectrum obtained by each of the transition. 1.b. i. "Nanomaterials are classified based on dimensions" mentions the types with suitable examples (4M) ii. What do you mean by Blue shift? In nanomaterials, engergy gap and blue shift are interelated. Illustrate with example (2M) 1.c. Comment on Scattering of light, Raman scattering, stokes lines and anti-stokes lines. (4M)

Archive page 7 from 10__1vFS6E1bKV.pdf; unit match candidate score 0.8333; source text/OCR may be partial. (4.0 Marks) (4.0 Marks) (5.0 Marks) 5.b. i. Write the reaction of synthesis of epoxy resine starting from Epichlorohydrin and Bisphenol A - 2M ii. Mention the curing agents added to epoxy resine, what is role played by it? - 2M 5.c. A polymer has the following molar mass distribution 1. 50 molecules of molecular weight 5000 g.mol 2. 75 molecules of molecular weight 6000 g.mole Calulate number average and weight average moleculare weight of the polymer. 4M 5.d. i. "PVC is more crystalline than PE" Justify the statement (2M) ii. Mention any three principles of Green Chemistry with example (3M) -1 -1

Archive page 5 from 11__1zfeA5C-J7.pdf; unit match candidate score 0.8333; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (5.0 Marks) 4.d. Why is viscometer placed in the water bath during flow rate measurements in viscosity avergage molecular mass determination experiment? 5.a. (i) Why is poly vinyl chloride more crystalline than polyethylene? (2 marks) (ii) How does the addition of triphenyl phosphate (Ph PO) to a non elastic polymer affect its flexibilty? Why? (2 marks) (iii) Is PMMA an addition or condensation polymer? Mention one of its application. (2 marks) 5.b. (i) Name the monomeric units from which epoxy resin is synthesised and give the reaction corresponding to the synthesis of epoxy resin from these monomers (reaction depicting the synthesis of monomeric units is not required). 3 Marks (ii) Which of the two monomeric units is used in excess during the synthesis of epoxy resin? Why? (2 Marks) (iii) Give one important application of epoxy resin. (1 Mark) 5.c. (i) Calculate the viscosity average molecular mass of the polymer sample which has 70 molecules of molecular mass 800, 100 molecles of molecular mass 2500 and 130 molecules of molecular mass 3500. Given the Mark Houwink coefficient in the viscosity measurements is 0.65. Mention of formula and final answer is required) (3 marks) (ii) As per the principles of Green Chemistry, why should derivatization be reduced during a chemical synthesis? (2 marks) 3 4

Archive page 5 from 1__1rBrYYqw3s.pdf; unit match candidate score 1.4167; source text/OCR may be partial. (6.0 Marks) (6.0 Marks) (6.0 Marks) (4.0 Marks) 4.a. i) Explain how the structure of polymer affects its crystallinity. ii) What are the monomers used in the synthesis of Butyl rubber ? iii) Write any two principles of Green chemistry. 4.b. i) Give the synthesis of kevlar from 1, 4 phenylene diamine. ii) "Kevlar is 5 times stronger than steel", justify the statement. iii) Give any one application and one disadvantage of kevlar. 4.c. Calculate the number average, weight average and viscosity average molecular weight of a polymer having 10 molecules of molecular weight 3000,

15 molecules of molecular weight 3500 and 25 molecules of molecular weight 4000. (Given : a = 0.65). 4.d. The optical properties of nano materials change with size of the nanopartcles. Explain using i) Quantum confinement ii) Surface plasmon resonance

Archive page 5 from 2__11pj9jsL0V.pdf; unit match candidate score 1.3333; source text/OCR may be partial. (6.0 Marks) (6.0 Marks) (6.0 Marks) (4.0 Marks) 4.a. Using acetone and hydrogen cyanide as the starting materials, outline the reactions involved in the synthesis of PMMA. Give any two applications of PMMA. 4.b. i) What are Biodegradable polymers? Explain with a suitable example. ii) Write the steps involved in the biodegradation of polymers. Give any two advantages. 4.c. What are Organic Light Emitting Diodes? Explain the construction, working principle, and applications of OLEDs. 4.d. Account for the following i) The strength of nanomaterials is greater than bulk material. ii) Design for Degradation and Catalysis. (Green Chemistry Principles)

Archive page 6 from 3__1MwY1CTl_j.pdf; unit match candidate score 1.1667; source text/OCR may be partial. 3/25/25, 3:51 PM ESA - Dec 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 6/7 (4.0 Marks) (6.0 Marks) (8.0 Marks) 4.a. Classify the nanomaterials based on their dimensions with suitable examples 4.b. (a) Write the differences between addition and condensation polymerization processes (2M) (b) Discuss the chemical synthesis of polymethyl methacrylate with proper reactants and reagents (4M) 4.c. (a) Discuss the construction and working principles of organic light emitting diodes (4M) (b) Describe the various doping processes of conducting polymers (4M)

Archive page 6 from 4__1eKMFaJ2lx.pdf; unit match candidate score 1.0; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 6/7 (6.0 Marks) (6.0 Marks) (6.0 Marks) 4.a. i) Explain how the structure of polymer affects its crystallinity. ii) What are the monomers used in the synthesis of Butyl rubber ? iii) Write any two principles of Green chemistry. 4.b. i) Give the synthesis of kevlar from 1, 4 phenylene diamine. ii) "Kevlar is 5 times stronger than steel", justify the statement. iii) Give any one application and one disadvantage of kevlar. 4.c. Calculate the number average, weight average and viscosity average molecular weight of a polymer having 10 molecules of molecular weight 3000, 15 molecules of molecular weight 3500 and 25 molecules of molecular weight 4000. (Given : a = 0.65).

Archive page 7 from 4__1eKMFaJ2lx.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 7/7 (4.0 Marks) (3.0 Marks) 4.d. The optical properties of nano materials change with size of the nanopartcles. Explain using i) Quantum confinement ii) Surface plasmon resonance 4.e. i) Define viscosity co-efficient. ii) Write expression of Lambert- Beer's law and explain the terms.

Archive page 6 from 5__1pIeTzocnU.pdf; unit match candidate score 0.9167; source text/OCR may be partial. 11/11/24, 11:34 AM ESA -JAN 2024 - UE23CY151A (set- 2) about:blank 6/7 Discuss the structure-property relationship with regard to the following properties of polymers (i) Stereoregularity (ii) Tensile strength (6.0 Marks) (6.0 Marks) (6.0 Marks) 4.a. 4.b. Give the synthesis of epoxy resin by the polymerization of monomers. How does curing impart a high degree of hardness to the polymer? Give any two applications of epoxy resin. 4.c. Complete the following table for OLEDs : Anode Cathode Emissive layer Conductive layer Advantage Application

Archive page 7 from 5__1pIeTzocnU.pdf; unit match candidate score 0.5; source text/OCR may be partial. 11/11/24, 11:34 AM ESA -JAN 2024 - UE23CY151A (set- 2) about:blank 7/7 (i) When the particle size is of nano order there is a decrease in the conductance of the species when compared to bulk materials. Give reasons. (ii)State any one principle of Green Chemistry and give relevant examples. (4.0 Marks) (3.0 Marks) 4.d. 4.e. 25 ml of sewage water sample is mixed with 25 ml of K Cr O solution and the unreacted K Cr O consumed 9 ml of 0.25N FAS to give the colour change with ferroin indicator. Under similar conditions, in blank titration 15ml of FAS is used up. Calculate the COD (Chemical Oxygen Demand) of the sample in mg/L (ppm).

Archive page 6 from 6__1H6QH_PVNf.pdf; unit match candidate score 1.0; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 6/7 (6.0 Marks) (6.0 Marks) (6.0 Marks) 4.a. i) Explain how the structure of polymer affects its crystallinity. ii) What are the monomers used in the synthesis of Butyl rubber ? iii) Write any two principles of Green chemistry. 4.b. i) Give the synthesis of kevlar from 1, 4 phenylene diamine. ii) "Kevlar is 5 times stronger than steel", justify the statement. iii) Give any one application and one disadvantage of kevlar. 4.c. Calculate the number average, weight average and viscosity average molecular weight of a polymer having 10 molecules of molecular weight 3000, 15 molecules of molecular weight 3500 and 25 molecules of molecular weight 4000. (Given : a = 0.65).

Archive page 7 from 6__1H6QH_PVNf.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 7/7 (4.0 Marks) (3.0 Marks) 4.d. The optical properties of nano materials change with size of the nanopartcles. Explain using i) Quantum confinement ii) Surface plasmon resonance 4.e. i) Define viscosity co-efficient. ii) Write expression of Lambert- Beer's law and explain the terms.

Archive page 5 from 7__16pEjcSTVO.pdf; unit match candidate score 1.0833; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (6.0 Marks) 4.d. Why should viscometer be placed in water bath during flow rate measurements in viscosity average molecular mass determination experiment. 5.a. (i) Define Glass transition temperature of a polymer? What does it indicate about the polymer. (ii) Give any three points of difference between addition and condensation polymer. 5.b. (i) Give the reaction of synthesis of Kevlar from the monomeric units. (ii) What is the reason for the high strength of kevlar (iii) Give one application of kevlar. 5.c. (i) Explain any one principle of Green Chemistry with suitable example. (ii) If a polymeric sample has 100 molecules of molecular mass 1000g/mol, 200 molecules of molecular mass 2000 g/mol and 500 molecules of molecular mass 5000g/mol. Calculate the number average and mass average molecular mass of the sample.

Archive page 6 from 8__1015pA2nzY.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/25/25, 4:03 PM ESA - Dec 2024 - UE22CY151B/UE22CY151A (set- 1) about:blank 6/7 (5.0 Marks) (6.0 Marks) (6.0 Marks) 4.d. Describe the process of tinning. Mention one advantage of tinning. 5.a. i)Distinguish between addition and condensation polymerization. ii)write the relevent reaction of Nylon 6,6. 5.b. Calculate the number average, weight average and viscosity average of a polymer having 10 molecules of molecular weight is 3000, 15 molecules of molecular weight is 3500 and 25 molecules of molecular weight is 4000. (Given a=0.65)

Archive page 7 from 8__1015pA2nzY.pdf; unit match candidate score 0.5833; source text/OCR may be partial. 3/25/25, 4:03 PM ESA - Dec 2024 - UE22CY151B/UE22CY151A (set- 1) about:blank 7/7 (6.0 Marks) (2.0 Marks) 5.c. i) Explain how the structure of polymer affects its crystallinity. ii) What are the monomers used in the synthesis of Butyl rubber? iii) Write any two principles of Green chemistry. 5.d. "Kevlar is 5 times stronger than steel" justify the statement.

Archive page 8 from 9__14fBQb1-hv.pdf; unit match candidate score 0.75; source text/OCR may be partial. 11/8/24, 1:02 PM ESA - DEC 2023 - UE22CY151B/UE22CY151A (set- 1) about:blank 8/9 (4.0 Marks) (4.0 Marks) (5.0 Marks) 5.b. i. Kevlar is very strong material – about 5 times stronger than steel- Justify the statement ii. Mention any two disadvantages of Kevlar 5.c. A polymer has the following molar mass distribution 1. 50 molecules of molecular weight 5000 g/mole 2. 75 molecules of molecular weight 6000 g/mole Calculate number average and weight average molecular weigt of the polyymer (4M) 5.d. i. Polyethylene(PE) is more crystalline then polyvinylacetate Justify the statement (2M) ii. Mention any three green chemistry Principles with example (2M)

Archive page 5 from Chem-EC[2025]__1ZtWpoG9Dv.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 2) about:blank 5/6 (6.0 Marks) (6.0 Marks) (6.0 Marks) 4.a. Account for the following- (i) Polyvinylchloride is more crystalline than polyethene. (2marks) (ii) During the synthesis of butyl rubber, 5-10% isoprene is used. (2marks) (iii) Bakelite and polypropylene differ completely in the way they respond to heat. (2marks) 4.b. Give a polymer as an example for the below mentioned applications - (i) A polymer used to impart shrinkage/crease resistance to fabrics (Ii) A polymer used in making seals for vacuum applications (iii) A polymer used in manufacture of aquariums and aircraft canopies. (iv) A polymer used in the conductive layer of an OLED (Organic Light Emitting Diode) (v) A polymer used in applications in prostheses, implants, tendon/ligament repair. (v) A polymer that is a component of personal armor such as combat helmets. 4.c. Account for the following (i) Bending of bulk Cu happens readily but nano sized Cu particles are super hard. (2marks) (ii) Gold spheres of $>$ 50 nm absorb at λ = 575 nm whereas gold spheres of 2-5 nm absorb at λ = 517 nm. (2marks) (iii) Bulk gold is catalytically inactive but gold nanoparticles are catalytically active. (2marks)

Archive page 6 from Chem-EC[2025]__1ZtWpoG9Dv.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 2) about:blank 6/6 (7.0 Marks) 4.d. (i) Why should the use of blocking group and unnecessary derivatization be avoided during a chemical reaction. (2 marks) (ii) Elaborate on 'Safer Synthesis' principle of 'Green Chemistry'. (2 marks) (iii) 4 g of Ferrous ammonium sulphate (FAS) is accurately weighed out, half a test tube conc. H SO is added to it and it is made upto mark in a 250 ml standard flask and taken in the burette. 25 ml of waste paper is mixed with acidified dichromate and this consumes 12.5 ml of the standard FAS for complete reaction in the estimation of COD of waste water sample. If the blank titre value when no waste water sample is taken is 45ml of the same standard FAS, Calculate the COD of one litre of water sample . Given the Gram equivalent weight of FAS is 392.13. (3 marks) 2 4

Archive page 6 from Chem-RR[2025]__1xN30tqzNj.pdf; unit match candidate score 1.25; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 1) about:blank 6/7 (6.0 Marks) (6.0 Marks) (6.0 Marks) 4.a. Account for, i) High strength of kevlar fibre.

(2M) ii) High electrical conductance of conducting polyaniline. (4M) Explain the above with suitable structures. 4.b. i)Give the synthesis of carbon fibre from polyacrylonitrile (with reactions). (4M) ii) Calculate the weight average molecular weight of a polymer having 10 molecules of molecular weight is 3000, 15 molecules of molecular weight is 3500 and 25 molecules of molecular weight is 4000. (2M) 4.c. i) Describe the construction and working of OLEDs. (4M) ii) Write any two principles of green chemistry. (2M)

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 2) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2c483373ae696940.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 2) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2c483373ae696940.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 2) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2c483373ae696940.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9960; only the JSON segmentation supplies the displayed timestamps.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 2) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2c483373ae696940.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 1) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2d33abc42be29800.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 1) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2d33abc42be29800.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 1) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2d33abc42be29800.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9969; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Nanomaterials: Introduction, classification, size-dependent properties: surface area (adsorption), surface energy (reactivity), optical properties - blue shift (Part 1) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__2d33abc42be29800.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__3ce99088727d358f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__3ce99088727d358f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__3ce99088727d358f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9970; only the JSON segmentation supplies the displayed timestamps.
Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__3ce99088727d358f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Polymers: Introduction, types of polymerizations, definition of glass transition temperature transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__6749a5eae673dbbc.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymers: Introduction, types of polymerizations, definition of glass transition temperature transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__6749a5eae673dbbc.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymers: Introduction, types of polymerizations, definition of glass transition temperature transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__6749a5eae673dbbc.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9986; only the JSON segmentation supplies the displayed timestamps.
Polymers: Introduction, types of polymerizations, definition of glass transition temperature transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__6749a5eae673dbbc.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Structure-property relationship transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/043_Structure-property_relationship__e79c2e83734020ae.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Structure-property relationship transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/043_Structure-property_relationship__e79c2e83734020ae.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Structure-property relationship transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/043_Structure-property_relationship__e79c2e83734020ae.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Structure-property relationship transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/043_Structure-property_relationship__e79c2e83734020ae.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Synthesis and properties of epoxy resin transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/045_Synthesis_and_properties_of_epoxy_resin__7c6d59e1128dc6e8.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Synthesis and properties of epoxy resin transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/045_Synthesis_and_properties_of_epoxy_resin__7c6d59e1128dc6e8.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Synthesis and properties of epoxy resin transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/045_Synthesis_and_properties_of_epoxy_resin__7c6d59e1128dc6e8.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Synthesis and properties of epoxy resin transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/045_Synthesis_and_properties_of_epoxy_resin__7c6d59e1128dc6e8.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 1) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_1__3d41b078ed8e32dd.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 1) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_1__3d41b078ed8e32dd.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 1) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_1__3d41b078ed8e32dd.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9971; only the JSON segmentation supplies the displayed timestamps.
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 1) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_1__3d41b078ed8e32dd.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_2__73d1b9e68bb92928.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_2__73d1b9e68bb92928.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_2__73d1b9e68bb92928.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9975; only the JSON segmentation supplies the displayed timestamps.
Polymer composites - synthesis, properties of Kevlar and Carbon fibers (Part 2)) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers_Part_2__73d1b9e68bb92928.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__9f6f7b970d3fb69d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__9f6f7b970d3fb69d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__9f6f7b970d3fb69d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9955; only the JSON segmentation supplies the displayed timestamps.
Average molecular weight of polymers - number average, weight average, and viscosity average molecular weight; numerical problems based on the same transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__9f6f7b970d3fb69d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Conducting polymers - examples, types of doping transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/048_Conducting_polymers_-_examples_types_of_doping__64decc463863d8a0.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Conducting polymers - examples, types of doping transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/048_Conducting_polymers_-_examples_types_of_doping__64decc463863d8a0.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Conducting polymers - examples, types of doping transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/048_Conducting_polymers_-_examples_types_of_doping__64decc463863d8a0.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Conducting polymers - examples, types of doping transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/048_Conducting_polymers_-_examples_types_of_doping__64decc463863d8a0.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Organic light-emitting diodes (OLEDs): Construction, working principle, and light emission mechanism transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__d329dc2c82435f24.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Organic light-emitting diodes (OLEDs): Construction, working principle, and light emission mechanism transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__d329dc2c82435f24.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Organic light-emitting diodes (OLEDs): Construction, working principle, and light emission mechanism transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__d329dc2c82435f24.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9971; only the JSON segmentation supplies the displayed timestamps.
Organic light-emitting diodes (OLEDs): Construction, working principle, and light emission mechanism transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__d329dc2c82435f24.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Green chemistry: Introduction, twelve principles of green chemistry with examples transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__65cb11f75f072aae.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Green chemistry: Introduction, twelve principles of green chemistry with examples transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__65cb11f75f072aae.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Green chemistry: Introduction, twelve principles of green chemistry with examples transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__65cb11f75f072aae.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9983; only the JSON segmentation supplies the displayed timestamps.
Green chemistry: Introduction, twelve principles of green chemistry with examples transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__65cb11f75f072aae.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Nanomaterials Introduction classification size dependent properties surface area adsorption surface energy rea (converted PDF; original hash ae5d11c83d7ee0572e21b2a94bbb3623e085fa917470c7d72dff1a365eb13c76)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/040_Nanomaterials_Introduction_classification_size-dependent_properties_surface_area_adsorption_surface_energy_rea__9542b34cc2143aa3.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Thermal properties melting point electrical properties and magnetic properties superparamagnetism Graphene s (converted PDF; original hash b05ecebc066c8c1af6db2b16f0d65865d70cbadf9ab0b7c9ab61c058936838cb)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__1c09b2e51011f2db.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Polymers Introduction types of polymerizations definition of glass transition temperature	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Polymers Introduction types of polymerizations definition of glass transition temperature	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Polymers Introduction types of polymerizations definition of glass transition temperature (converted PDF; original hash 1ad6fc41336a211ac5c12645b0d6b7c0ef8d13c9f0d4bdd2502112a3c03b4f12)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__fb4b07d3ff4fffb9.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Structure property relationship	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Structure property relationship	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Structure property relationship (converted PDF; original hash 93190733cb7dd6e8fc21d21da279c07476ce73574251c19ef48cef845c3a81cd)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/043_Structure-property_relationship__ac30080565c6d4d9.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Commercial polymers synthesis and properties of Plexiglass	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Commercial polymers synthesis and properties of Plexiglass	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Commercial polymers synthesis and properties of Plexiglass (converted PDF; original hash 57bca0020e382c0b12e473ceb72e9250bfa79c292485bd1c19689ef1349466906)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/044_Commercial_polymers_synthesis_and_properties_of_Plexiglass__ebc28a1781fd1fb2.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Synthesis and properties of epoxy resin	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Synthesis and properties of epoxy resin	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Synthesis and properties of epoxy resin (converted PDF; original hash 19fbeebb19cb44c83573666f8878a6a091c1664eec36ee1544b0d9df00364b9d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/045_Synthesis_and_properties_of_epoxy_resin__dfd57a34168c7f3e.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Polymer composites synthesis properties of Kevlar and Carbon fibers	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Polymer composites synthesis properties of Kevlar and Carbon fibers	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Polymer composites synthesis properties of Kevlar and Carbon fibers (converted PDF; original hash 2560c8f6143cbf4231c5c6dfa9794b01a13c27aa4e20369f322f680e0b94a93b)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/046_Polymer_composites_-_synthesis_properties_of_Kevlar_and_Carbon_fibers__d89075783b33c052.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Average molecular weight of polymers number average weight average and viscosity average molecular weight nu	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Average molecular weight of polymers number average weight average and viscosity average molecular weight nu	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Average molecular weight of polymers number average weight average and viscosity average molecular weight nu (converted PDF; original hash ae4d397b497d8ca2e63777121805fdc0902e941301d0376f18ab85c8238855d1)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/047_Average_molecular_weight_of_polymers_-_number_average_weight_average_and_viscosity_average_molecular_weight_nu__57e6e9198a641301.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Conducting polymers examples types of doping	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Conducting polymers examples types of doping	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Conducting polymers examples types of doping (converted PDF; original hash ca326f637ca66e6900f3cf0d5fd9e089f7a5e25ff2ec2719b20b860aa87aa94f)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/048_Conducting_polymers_-_examples_types_of_doping__cf4e1fdbc1c9b41c.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Organic light emitting diodes OLEDs Construction working principle and light emission mechanism	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Organic light emitting diodes OLEDs Construction working principle and light emission mechanism	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Organic light emitting diodes OLEDs Construction working principle and light emission mechanism (converted PDF; original hash a9b807e70297b390226c96b5b111ec2155b7d8d040bb5fb99cf1fc594aa8b61d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/049_Organic_light-emitting_diodes_OLEDs_Construction_working_principle_and_light_emission_mechanism__82258b2fc2e2ddf9.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Green chemistry Introduction twelve principles of green chemistry with examples	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Green chemistry Introduction twelve principles of green chemistry with examples	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Green chemistry Introduction twelve principles of green chemistry with examples (converted PDF; original hash ec2cb5da0a96641aa41cc7de99dfe13473d90ebceb1004effb11919c74e3caf1)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/050_Green_chemistry_Introduction_twelve_principles_of_green_chemistry_with_examples__43c3c6adb38e0a00.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Experiential learning Based on nanomaterials	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/051_Experiential_learning-_Based_on_nanomaterials__de26aa31ff70371c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Experiential learning Based on nanomaterials	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/051_Experiential_learning-_Based_on_nanomaterials__de26aa31ff70371c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Experiential learning Based on nanomaterials (converted PDF; original hash a633a07de184f9133999395e1afcbe5124c4399b4824bea6778640e31355d34b)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/Slides/051_Experiential_learning-_Based_on_nanomaterials__de26aa31ff70371c.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Extracted historical pages: Chem_Extra_Topics__1-HfOeZNwo.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Extra/Chem_Extra_Topics__1-HfOeZNwo.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 10__1vFS6E1bKV.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/10__1vFS6E1bKV.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 11__1zfeA5C-J7.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/11__1zfeA5C-J7.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 1__1rBrYYqw3s.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/1__1rBrYYqw3s.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 2__11pj9jsL0V.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/2__11pj9jsL0V.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 3__1MwY1CTl_j.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/3__1MwY1CTl_j.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Source	Path	Use
Extracted historical pages: 4__1eKMFaJ2lx.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/4__1eKMFaJ2lx.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 5__1pIeTzocnU.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/5__1pIeTzocnU.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 6__1H6QH_PVNf.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/6__1H6QH_PVNf.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 7__16pEjcSTVO.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/7__16pEjcSTVO.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 8__1015pA2nzY.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/8__1015pA2nzY.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 9__14fBQb1-hv.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/9__14fBQb1-hv.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: Chem-EC[2025]__1ZtWpoG9Dv.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2025/Chem-EC[2025]__1ZtWpoG9Dv.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: Chem-RR[2025]__1xN30tqzNj.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2025/Chem-RR[2025]__1xN30tqzNj.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.

2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.
SRC00002	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00003	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00004	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00005	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00006	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00007	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00008	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00009	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00010	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00011	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00012	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00013	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00014	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00015	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0058. Sidecar variants compare...
SRC00016	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00017	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0058. Sidecar variants compare...
SRC00018	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00019	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00020	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00021	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00022	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0046. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00023	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0046. Sidecar variants compare...
SRC00024	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0046. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00025	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0046. Sidecar variants compare...
SRC00026	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00027	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0056. Sidecar variants compare...
SRC00028	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00029	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0056. Sidecar variants compare...
SRC00030	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00031	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...
SRC00032	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00033	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...
SRC00034	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00035	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00036	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00037	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00038	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00039	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00040	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00041	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00042	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00043	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00044	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00045	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00046	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00047	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00048	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00049	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00050	0 covered · 0 excluded · 14 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00052	0 covered · 0 excluded · 14 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00051	0 covered · 0 excluded · 14 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00053	0 covered · 0 excluded · 21 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00055	0 covered · 0 excluded · 21 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00054	0 covered · 0 excluded · 21 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00056	0 covered · 0 excluded · 18 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00058	0 covered · 0 excluded · 18 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00057	0 covered · 0 excluded · 18 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00059	0 covered · 0 excluded · 16 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00061	0 covered · 0 excluded · 16 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00060	0 covered · 0 excluded · 16 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00062	6 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00064	6 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00063	6 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00065	0 covered · 0 excluded · 8 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00067	0 covered · 0 excluded · 8 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00066	0 covered · 0 excluded · 8 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00068	0 covered · 0 excluded · 18 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00070	0 covered · 0 excluded · 18 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00069	0 covered · 0 excluded · 18 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00071	0 covered · 0 excluded · 14 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00073	0 covered · 0 excluded · 14 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00072	0 covered · 0 excluded · 14 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00074	0 covered · 0 excluded · 13 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00076	0 covered · 0 excluded · 13 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00075	0 covered · 0 excluded · 13 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00077	0 covered · 0 excluded · 14 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00079	0 covered · 0 excluded · 14 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00078	0 covered · 0 excluded · 14 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00080	0 covered · 0 excluded · 24 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00082	0 covered · 0 excluded · 24 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00081	0 covered · 0 excluded · 24 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00083	3 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00085	3 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00084	3 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00086	1 covered · 0 excluded · 0 gaps	q46: Original archive PDF page 46; candidate unit association is text-based and unconfirmed.
SRC00087	2 covered · 0 excluded · 0 gaps	q1: Original archive PDF page 1; candidate unit association is text-based and unconfirmed. q7: Original archive PDF page 7; candidate unit association is text-based and unconfirmed.
SRC00088	1 covered · 0 excluded · 0 gaps	q5: Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
SRC00089	1 covered · 0 excluded · 0 gaps	q5: Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
SRC00090	1 covered · 0 excluded · 0 gaps	q5: Original archive PDF page 5; candidate unit association is text-based and unconfirmed.

Source	Unit counts	Example decisions
SRC00091	1 covered · 0 excluded · 0 gaps	q6: Original archive PDF page 6; candidate unit association is text-based and unconfirmed.
SRC00092	2 covered · 0 excluded · 0 gaps	q6: Original archive PDF page 6; candidate unit association is text-based and unconfirmed. q7: Original archive PDF page 7; candidate unit association is text-based and unconfirmed.
SRC00093	2 covered · 0 excluded · 0 gaps	q6: Original archive PDF page 6; candidate unit association is text-based and unconfirmed. q7: Original archive PDF page 7; candidate unit association is text-based and unconfirmed.
SRC00094	2 covered · 0 excluded · 0 gaps	q6: Original archive PDF page 6; candidate unit association is text-based and unconfirmed. q7: Original archive PDF page 7; candidate unit association is text-based and unconfirmed.
SRC00095	1 covered · 0 excluded · 0 gaps	q5: Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
SRC00096	2 covered · 0 excluded · 0 gaps	q6: Original archive PDF page 6; candidate unit association is text-based and unconfirmed. q7: Original archive PDF page 7; candidate unit association is text-based and unconfirmed.

Source	Unit counts	Example decisions
SRC00097	1 covered · 0 excluded · 0 gaps	q8: Original archive PDF page 8; candidate unit association is text-based and unconfirmed.
SRC00098	2 covered · 0 excluded · 0 gaps	q5: Original archive PDF page 5; candidate unit association is text-based and unconfirmed. q6: Original archive PDF page 6; candidate unit association is text-based and unconfirmed.
SRC00099	1 covered · 0 excluded · 0 gaps	q6: Original archive PDF page 6; candidate unit association is text-based and unconfirmed.

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